# Chapter 13: Reactions Equilibria in Systems Containing Components in Condensed Solution Part II

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# Outlines

# 13.5 Phase Stability Diagrams

- ✓ In this section, we introduce *phase stability diagrams* (also called *predominance diagrams*). These are diagrams which are used in multicomponent, multiphase reacting systems to display regions in thermodynamic variable space where various phases in the system are likely to be found.

- ✓ Consider equilibrium in the ternary system Si–C–O at 1000 ℃. The solid phases which can exist in this system are Si, $SiO_2$, SiC, and C, and the gas phase is a mixture of CO and $CO_2$.

- ✓ Just as fixing the activity of one component in a binary system fixes the activity of the other, the fixing of the activities of two of the components in a ternary system fixes the activity of the third.

# 13.5 Phase Stability Diagrams

- ✓ Thus, when the activities of C and $O_2$ are fixed in the system Si–C–O, the activity of Si is fixed and a definite equilibrium state exists.
- ✓ Thus, two-dimensional representation of the phase stability can be considered in the following ways:

  1. At constant temperature with $a_{\mathrm{C}}$ and $p_{\mathrm{O_2}}$ as the variables
  2. At constant $a_{\mathrm{C}}$(or constant $p_{\mathrm{O_2}}$) with $T$ and $p_{\mathrm{O_2}}$(or $a_{\mathrm{C}}$) as the variables

- ✓ Application of the phase rule to a three-component system indicates that an equilibrium among five phases has no degrees of freedom.

# 13.5 Phase Stability Diagrams

✓ Thus, if a gas phase is always present,

- Four condensed phases can be in equilibrium with one another and a gas phase at an invariant state (***F*** = 0 = *C* + 2 – Φ = 3 + 2 – 5).
- Three condensed phases can be in equilibrium with one another and a gas phase at an arbitrarily chosen temperature (***F*** = 1 = *C* + 2 – Φ = 3 + 2 – 4).
- Two condensed phases can be in equilibrium with one another and a gas phase at an arbitrarily chosen temperature and an arbitrarily chosen value of $a_{\mathrm{C}}$ or $p_{\mathrm{O}_2}$ (***F*** = 2 = *C* + 2 – Φ = 3 + 2 – 3).
- One condensed phase can be in equilibrium with the gas phase at an arbitrarily chosen temperature and arbitrarily chosen values of $a_{\mathrm{C}}$ and $p_{\mathrm{O}_2}$(***F*** = 3 = *C* + 2 – Φ = 3 + 2 – 2).

# 13.5 Phase Stability Diagrams

- ✓ Since four solid phases can exist, there are
  - Four possible equilibria involving three condensed phases and a gas phase (found by taking four solid phases three at a time, or $4!/(3!\cdot 1!)=4$)
  - Six possible equilibria involving two condensed phases and a gas phase (found by taking four solid phases two at a time, or $4!/(2!\cdot 2!)=6$)
  - Four possible equilibria involving one condensed phase and the gas phase (found by taking four solid phases one at a time, or $4!/(1!\cdot 3!)=4$)
- ✓ Consider construction of the phase stability diagram for the system Si–C–O at 1000 ℃ using $\log a_{\mathrm{C}}$ and $\log a_{\mathrm{C}}$ as the variables.

# 13.5 Phase Stability Diagrams

✓ As shown, there are six possible equilibria involving two condensed phases and a gas phase: namely,

1. Si–$SiO_2$–gas
2. Si–SiC–gas
3. SiC–$SiO_2$–gas
4. SiC–C–gas
5. $SiO_2$–C–gas
6. Si–C–gas

✓ And the four possible equilibria involving three condensed phases and a gas phase are

1. Si–$SiO_2$–SiC–gas
2. Si–SiC–C–gas
3. $SiO_2$–SiC–C–gas
4. Si–$SiO_2$–C–gas

# 13.5 Phase Stability Diagrams

✓ Consider the equilibria between two condensed phases and the gas phase.

**1. The equilibrium Si–$SiO_2$–gas phase**

For the reaction

$$\mathrm{Si}_{(s)} + \mathrm{O}_{2(g)} = \mathrm{SiO}_{2(s)} \qquad \text{(i)}$$

$$\Delta G^{\circ}_{(\mathrm{i})1273\ \mathrm{K}} = -683,400\ \mathrm{J} = -RT\ln\frac{1}{p_{\mathrm{O}_2}} = 8.3144\times1273\times2.303\log p_{\mathrm{O}_2}$$

or $\log p_{\mathrm{O}_2} = -28.04$ . Thus, at 1273 K, the equilibrium between Si and $SiO_2$ requires $\log p_{\mathrm{O}_2} = -28.04$, and this equilibrium, which is drawn as line *AB* in Figure 13.5a, is independent of $a_{\mathrm{C}}$. At lower values of $p_{\mathrm{O}_2}$ (the abscissa), Si is stable relative to $SiO_2$ and, at higher values, $SiO_2$ is stable relative to Si.

# 13.5 Phase Stability Diagrams

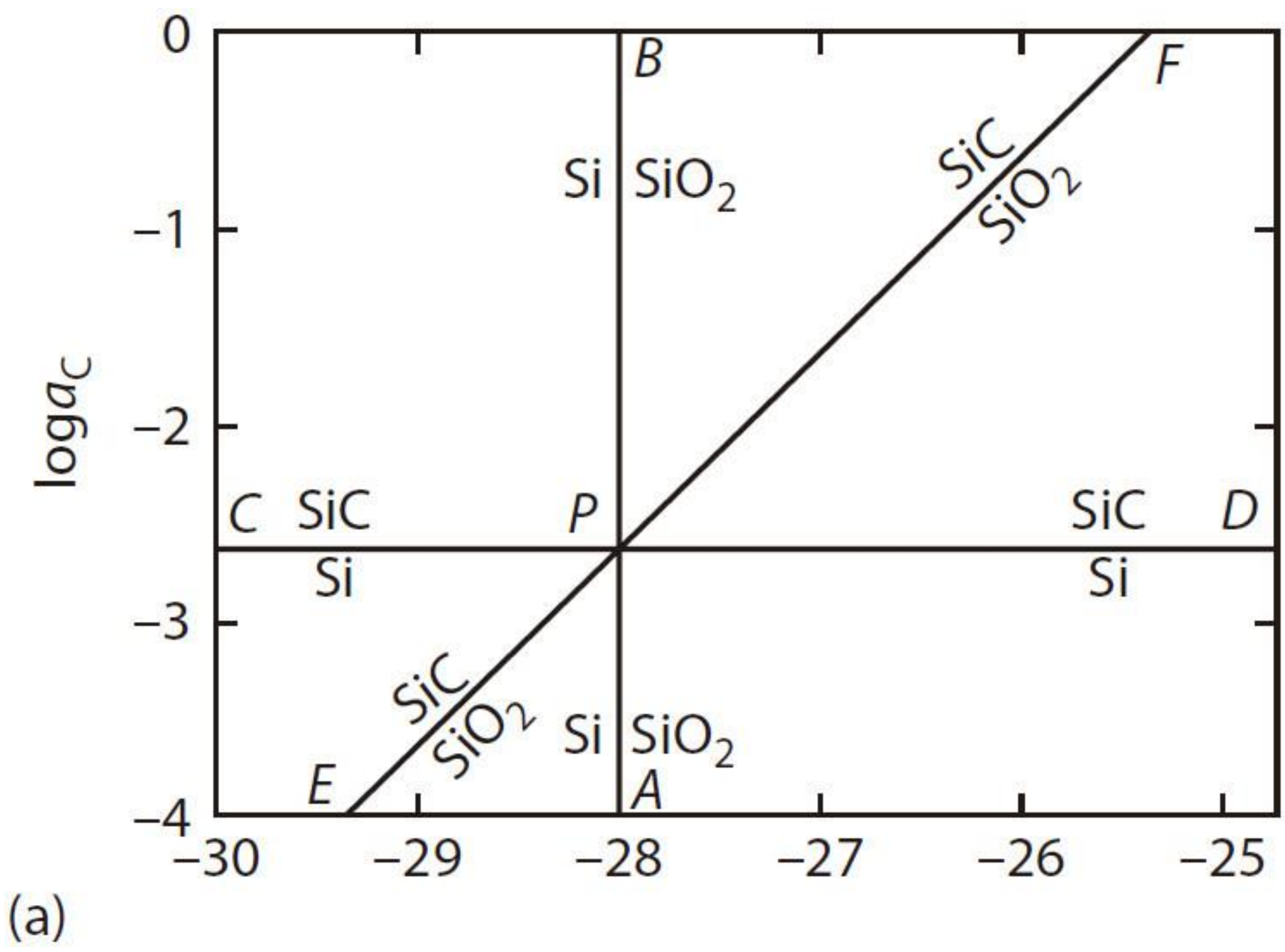


**Figure 13.5** (a) Construction of the phase stability diagram for the system Si–C–O at 1273 K.

**2. The equilibrium Si–SiC–gas phase**

For the reaction

$$\mathrm{Si}_{(s)} + \mathrm{C}_{(s)} = \mathrm{SiC}_{(s)} \tag{ii}$$

$$\Delta G^{\circ}_{\text{(ii)}1273\ \mathrm{K}} = -63,300\ \mathrm{J} = -RT\ln\frac{1}{a_{\mathrm{C}}} = 8.3144\times1273\times2.303\log a_{\mathrm{C}}$$

or $\log a_{\mathrm{C}} = -2.60$ . Thus, at 1273 K, the equilibrium between Si and SiC requires $\log a_{\mathrm{C}} = -2.60$, and this equilibrium, which is drawn as line *CD* in Figure 13.5a, is independent of $p_{\mathrm{O}_2}$ . At lower values of $a_{\mathrm{C}}$ , Si is relative to SiC and, at higher values of a C , SiC is stable relative to Si.

**3. The equilibrium SiC–SiO$_2$–gas phase**

Lines *AB* and *CD* intersect at the point *P* ($\log p_{O_2} = -28.04, \log a_C = -2.60$), which is the unique state at which the three solid phases Si, SiC, and SiO$_2$ are in equilibrium with one another and with a gas phase at 1273 K. The variation of $\log a_C$ with $\log p_{O_2}$ required for the equilibrium SiC–SiO$_2$ must pass through this point.

Combination of Equations (i) and (ii) gives

$$\mathrm{SiC}_{(s)} + \mathrm{O}_{2(g)} = \mathrm{SiO}_{2(s)} + \mathrm{C}_{(s)} \qquad \text{(iii)}$$

for which

$$\Delta G^{\circ}_{\mathrm{(iii)1273\,K}} = -620{,}100\ \mathrm{J} = -8.3144 \times 1273 \times 2.303 \log \frac{a_C}{p_{O_2}}$$

or

$$\log a_C = \log p_{O_2} + 25.44$$

which is drawn as line *EF* in Figure 13.5a. In states above this line, SiC is stable relative to SiO$_2$, and below the line, SiO$_2$ is stable relative to SiC.

## 4. The equilibrium SiC–C–gas phase

The equilibrium between solid SiC and solid C is a phase equilibrium which exists only at $a_{\mathrm{C}} = 1$, or $\log a_{\mathrm{C}} = 0$. Thus, in Figure 13.5a, the equilibrium between SiC and C exists along the $\log a_{\mathrm{C}} = 0$ line, at values of $\log p_{\mathrm{O_2}}$ less than –25.44, the point of intersection of line *EF* with the $\log a_{\mathrm{C}} = 0$ line.

## 5. The equilibrium $SiO_2$–C–gas phase

As with the equilibrium between SiC and C, the phase equilibrium between $SiO_2$ and C require $\log a_{\mathrm{C}} = 1$, and thus occurs along the $\log a_{\mathrm{C}} = 0$ line at values of $\log p_{\mathrm{O_2}}$ greater than –25.44.

**6. The equilibrium Si–C–gas phase**

The standard Gibbs free energy change at 1273 K for the reaction

$$\mathrm{Si}_{(s)} + \mathrm{C}_{(s)} = \mathrm{SiC}_{(s)}$$

is $\Delta G^{\circ}_{\mathrm{(ii)1273\ K}} = -63,300\ \mathrm{J}$, which, being negative, indicates that Si and C spontaneously react with one another to form SiC until either the C or the Si is consumed. Thus, Si and C cannot be in equilibrium with one another.

If the molar ratio Si/C in the system is greater than 1, the C is consumed by the reaction and equilibrium between the product SiC and the remaining Si is attained, and if the ratio is less than 1, the Si is consumed by the reaction and an equilibrium is attained between the SiC produced and the remaining C.

# 13.5 Phase Stability Diagrams

- ✓ The solid carbon phase exists only along the $\log a_{\mathrm{C}} = 1$ line, and thus, Figure 13.5a contains fields of stability of the single phases Si, $SiO_2$, and SiC.
- ✓ Consequently, of the six lines in the diagram radiating from point *P*, three represent stable equilibria involving two condensed phases and a gas phase, and three represent metastable equilibria involving two condensed phases and a gas phase.
- ✓ The problem is how to distinguish between the two types of equilibria. It is a property of such diagrams that the lines of metastable and stable equilibria radiate alternatively from a point such as *P* (see, for example, Figures 7.7 and 7.8a). Thus, one set of lines is *PA*-*PC*-*PF* and the other is *PE*-*PB*-*PD*.

# 13.5 Phase Stability Diagrams

- ✓ In Figure 13.5a, Si is stable relative to $SiO_2$ in states to the left of *PA* and is stable relative to SiC in states below *PC*. Thus, the line *PE* represents the metastable equilibrium between SiC and $SiO_2$.
- ✓ This identifies the stable equilibrium lines as being *PA*-*PC*-*PF* and, as shown in Figure 13.5b, defines the fields of stability of a single condensed phase with a gas phase, which, at constant temperature, have two degrees of freedom.
- ✓ These fields meet at lines which represent the equilibrium between two condensed phases and a gas phase (which, at constant temperature, have one degree of freedom), and the lines meet at points representing equilibrium among three condensed phases and a gas phase.

# 13.5 Phase Stability Diagrams

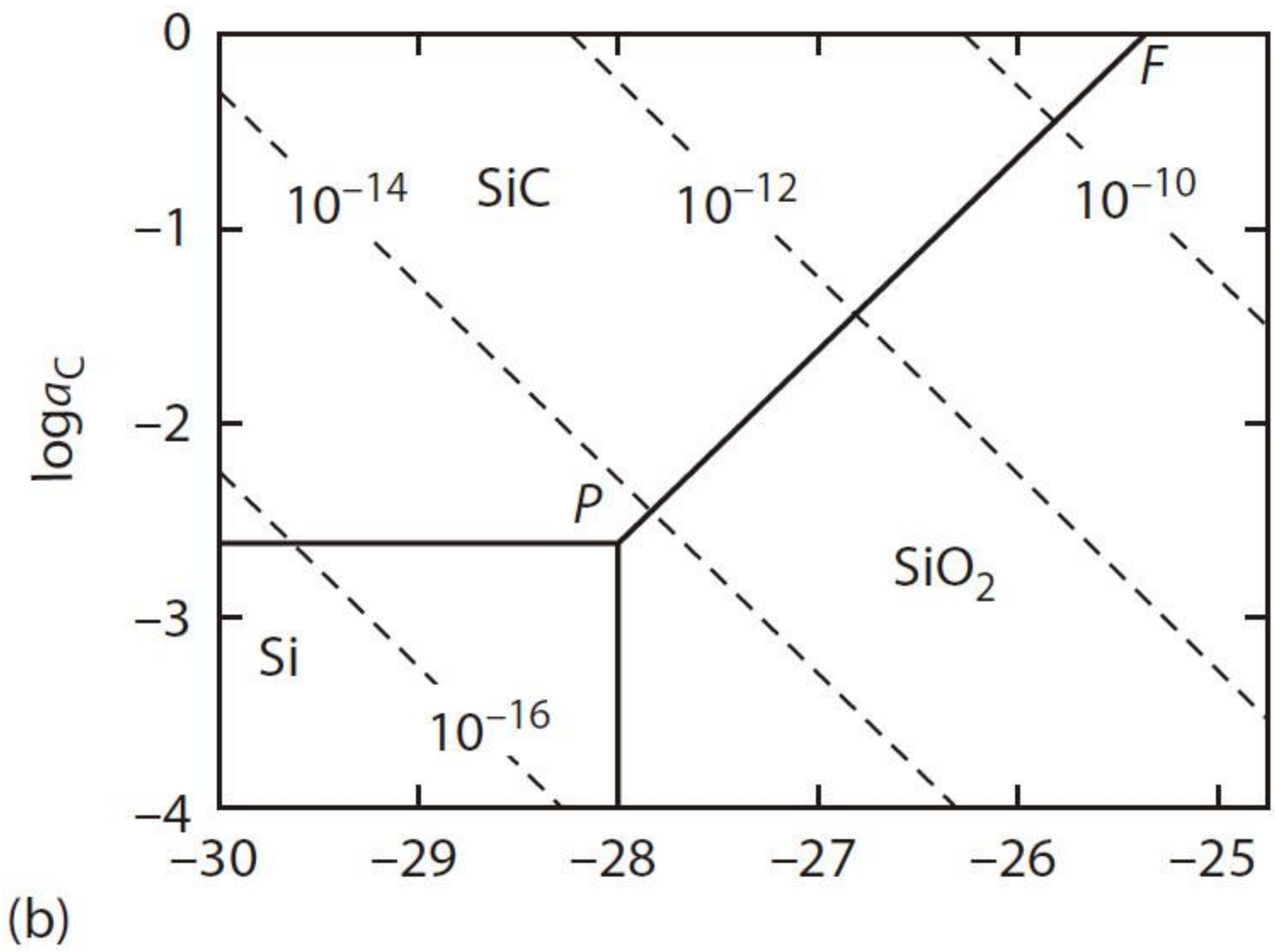


**Figure 13.5** *(Continued)* (b) $CO_2$ isobars in the phase stability diagram for Si–C–O at 1273 K.

# 13.5 Phase Stability Diagrams

- ✓ The point *P* is the state of equilibrium of Si, SiC, $SiO_2$, and a gas phase, and the point *F* is the state of equilibrium of SiC, $SiO_2$, C, and a gas phase.
- ✓ The partial pressures of CO and $CO_2$ are determined by the activities of carbon and oxygen in the system, and the iso-$p_{\mathrm{CO_2}}$ and iso-$p_{\mathrm{CO}}$ lines can be placed on the stability diagram as follows.
- ✓ For the reaction

$$\mathrm{C}_{(s)} + \mathrm{O}_{2(g)} = \mathrm{CO}_{2(g)} \qquad \text{(iv)}$$

$$\Delta G^{\circ}_{\text{(iv)}1273\text{ K}} = -395,200 \text{ J} = -8.3144 \times 1273 \times 2.303 \log \frac{p_{\mathrm{CO_2}}}{a_{\mathrm{C}}\, p_{\mathrm{O_2}}}$$

# 13.5 Phase Stability Diagrams

✓ Therefore,

$$16.21 = \log p_{CO_2} - \log a_C - \log p_{O_2}$$

or

$$\log a_C = -\log p_{O_2} - 16.21 + \log p_{CO_2}$$

✓ Thus, as shown in Figure 13.5b, the $CO_2$ isobars in the phase stability diagram are lines with slopes of –1.

✓ Similarly, for

$$C_{(s)} + \frac{1}{2}O_{2(g)} = CO_{(g)} \qquad \text{(v)}$$

$$\Delta G^{\circ}_{(v)1273\text{ K}} = -223,300\text{ J} = -8.3144 \times 1273 \times 2.303 \log \frac{p_{CO}}{a_C p_{O_2}^{1/2}}$$

or

$$\log a_C = -\frac{1}{2}\log p_{O_2} - 9.16 + \log p_{CO}$$

and, as shown in Figure 13.5c, the CO isobars are lines with slopes of $-\frac{1}{2}$.

# 13.5 Phase Stability Diagrams

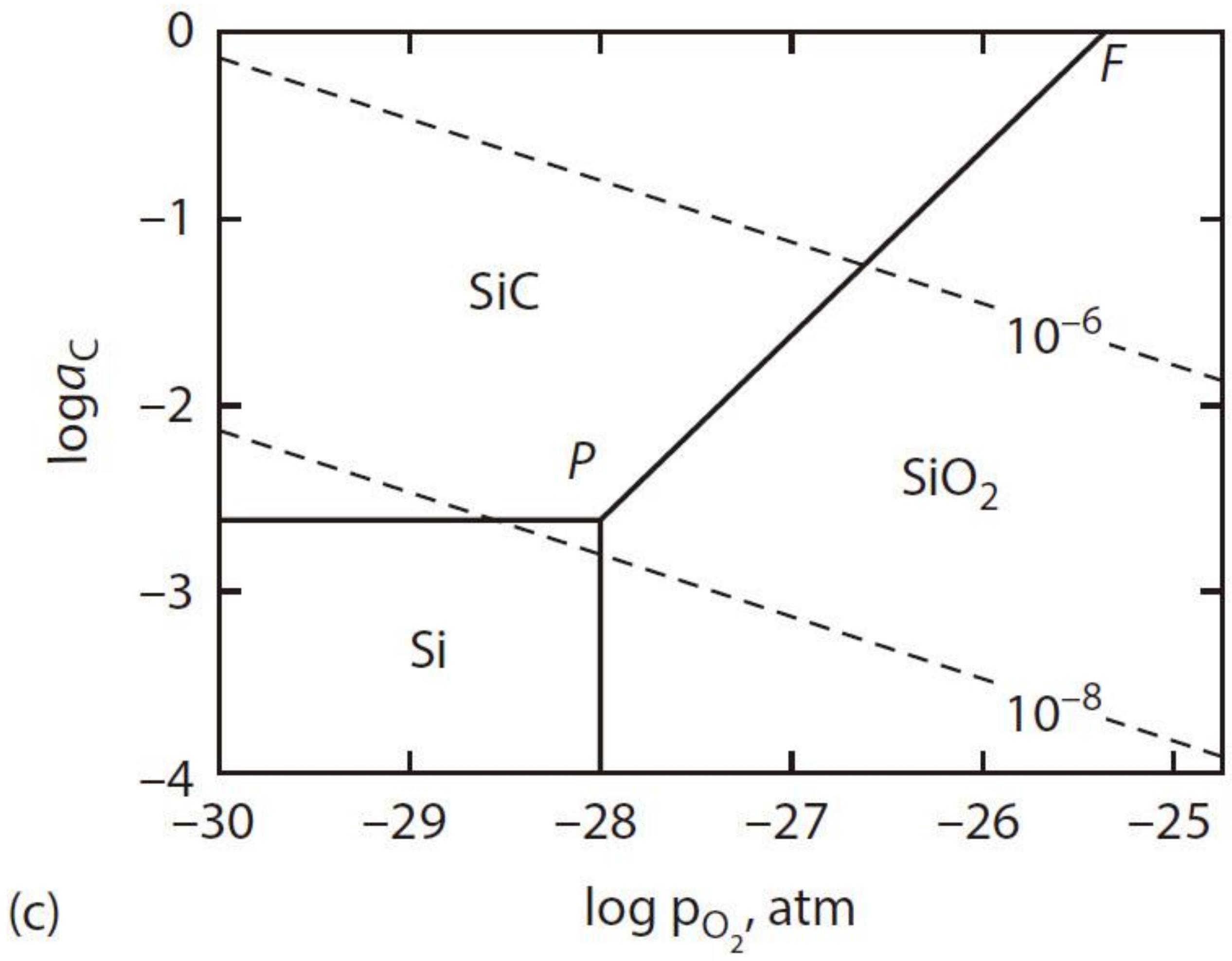


**Figure 13.5** *(Continued)* (c) CO isobars in the phase stability diagram for Si–C–O at 1273 K.

- ✓ The gaseous species SiO also occurs in the system Si–C–O, and SiO isobars can be drawn on the stability diagram.
- ✓ Since the partial pressure of SiO is determined by the activity of Si and the partial pressure of oxygen, the individual single condensed-phase stability fields have to be considered separately, and each equilibrium must involve the condensed phase of interest, SiO gas, and C and/or $O_2$.
- ✓ In the field of stability of Si, the equilibrium is

$$\mathrm{Si}_{(s)} + \frac{1}{2}\mathrm{O}_{2(g)} = \mathrm{SiO}_{(g)} \qquad \text{(vi)}$$

for which

$$\Delta G^{\circ}_{(\mathrm{vi})1273\,\mathrm{K}} = -209,200\ \mathrm{J} = -8.3144 \times 1273 \times 2.303 \log \frac{p_{\mathrm{SiO}}}{p_{\mathrm{O_2}}^{1/2}}$$

or

$$\log p_{\mathrm{SiO}} = \frac{1}{2}\log p_{\mathrm{O_2}} + 8.59$$

- ✓ Thus, as shown in Figure 13.5d, the SiO isobars in the stability field of Si are vertical lines, and $p_{\mathrm{SiO}}$ increases with increasing $p_{\mathrm{O_2}}$.

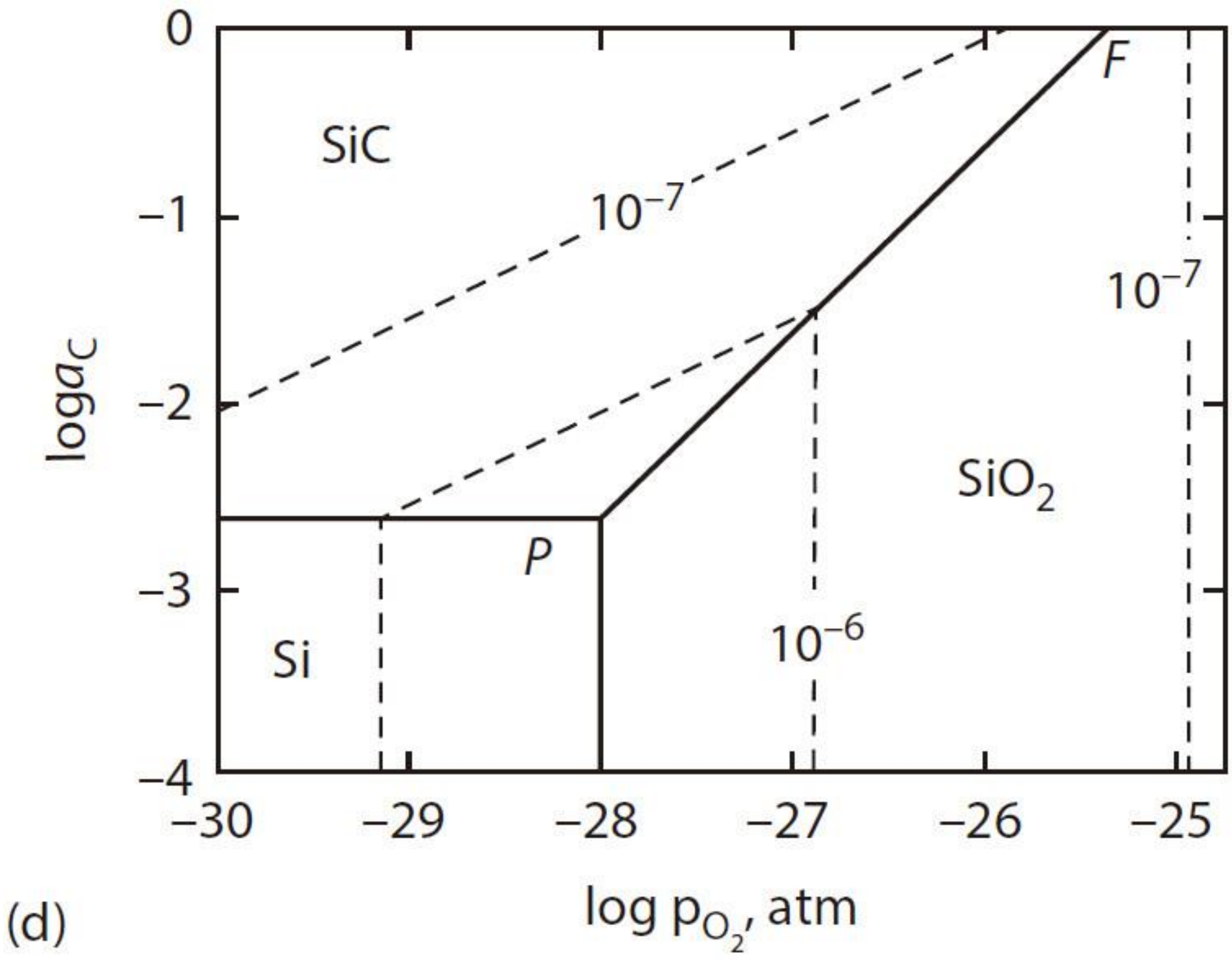


**Figure 13.5** *(Continued)* (d) SiO isobars in the phase stability diagram for Si–C–O at 1273 K.

# 13.5 Phase Stability Diagrams

✓ In the $SiO_2$ field, the equilibrium is

$$\mathrm{SiO}_{2(s)} = \mathrm{SiO}_{(g)} + \frac{1}{2}\mathrm{O}_{2(g)} \qquad \text{(vii)}$$

for which combination of $\Delta G^\circ_{\mathrm{(vi)1273\,K}}$ and $\Delta G^\circ_{\mathrm{(i)1273\,K}}$ gives

$$\Delta G^\circ_{\mathrm{(vii)1273\,K}} = -474,200\ \mathrm{J} = -8.3144 \times 1273 \times 2.303 \log\left(p_{\mathrm{SiO}}\, p_{\mathrm{O_2}}^{1/2}\right)$$

which gives

$$\log p_{\mathrm{SiO}} = -\frac{1}{2}\log p_{\mathrm{O_2}} - 19.45$$

✓ Thus, as shown in Figure 13.5d, the SiO isobars in the field of stability of $SiO_2$ are vertical lines and $p_{\mathrm{SiO}}$ decreases with increasing $p_{\mathrm{O_2}}$.

# 13.5 Phase Stability Diagrams

- ✓ In the field of stability of SiC, the equilibrium is

$$\mathrm{SiC}_{(s)} + \frac{1}{2}\mathrm{O}_{2(g)} = \mathrm{SiO}_{(g)} + \mathrm{C}_{(s)} \qquad \text{(viii)}$$

for which combination of $\Delta G^{\circ}_{(\text{vi})1273\text{ K}}$ and $\Delta G^{\circ}_{(\text{ii})1273\text{ K}}$ gives

$$\Delta G^{\circ}_{(\text{viii})1273\text{ K}} = -145,900\text{ J} = -8.3144\times 1273\times 2.303\log\frac{p_{\mathrm{SiO}}\cdot a_{\mathrm{C}}}{p_{\mathrm{O}_2}^{1/2}}$$

which gives

$$\log a_{\mathrm{C}} = \frac{1}{2}\log p_{\mathrm{O}_2} - \log p_{\mathrm{SiO}} + 5.99$$

- ✓ Thus, as shown in Figure 13.5d, the SiO isobars in the field of stability of SiC are lines with slopes of $1/2$ . At any temperature, the maximum value of $p_{\mathrm{SiO}}$ occurs in states in which Si and $SiO_2$ are in equilibrium with a gas phase.

# 13.5 Phase Stability Diagrams

✓ The activities of carbon and oxygen used as the variables in the construction of Figure 13.5 are determined by the individual values of $p_{CO}$ and $p_{CO_2}$ , which establish the equilibria

$$CO + \frac{1}{2}O_2 = CO_2$$

and

$$C + CO_2 = 2CO$$

and thus, the phase stability diagram can be constructed using $p_{CO}$ and $p_{CO_2}$ as the variables.

# 13.5 Phase Stability Diagrams

- ✓ The equilibrium corresponding to Equation (i) is

$$Si_{(s)} + 2CO_{2(g)} = SiO_{2(s)} + 2CO_{(g)} \qquad \text{(ix)}$$

- ✓ Combination of the Gibbs free energy changes for the reactions given by Equations (i), (iv), and (v) gives

$$\Delta G^{\circ}_{(\text{ix})1273\text{ K}} = -339,600 \text{ J} = -8.3144 \times 1273 \times 2.303 \log \frac{p_{CO}^2}{p_{CO_2}^2}$$

which gives

$$\log p_{CO} = \log p_{CO_2} + 6.97$$

- ✓ This is drawn as the line *AB* in Figure 13.6a (and corresponds to the line *AB* in Figure 13.5a). Above the line, Si is stable relative to $SiO_2$, and below the line, $SiO_2$ is stable relative to Si.

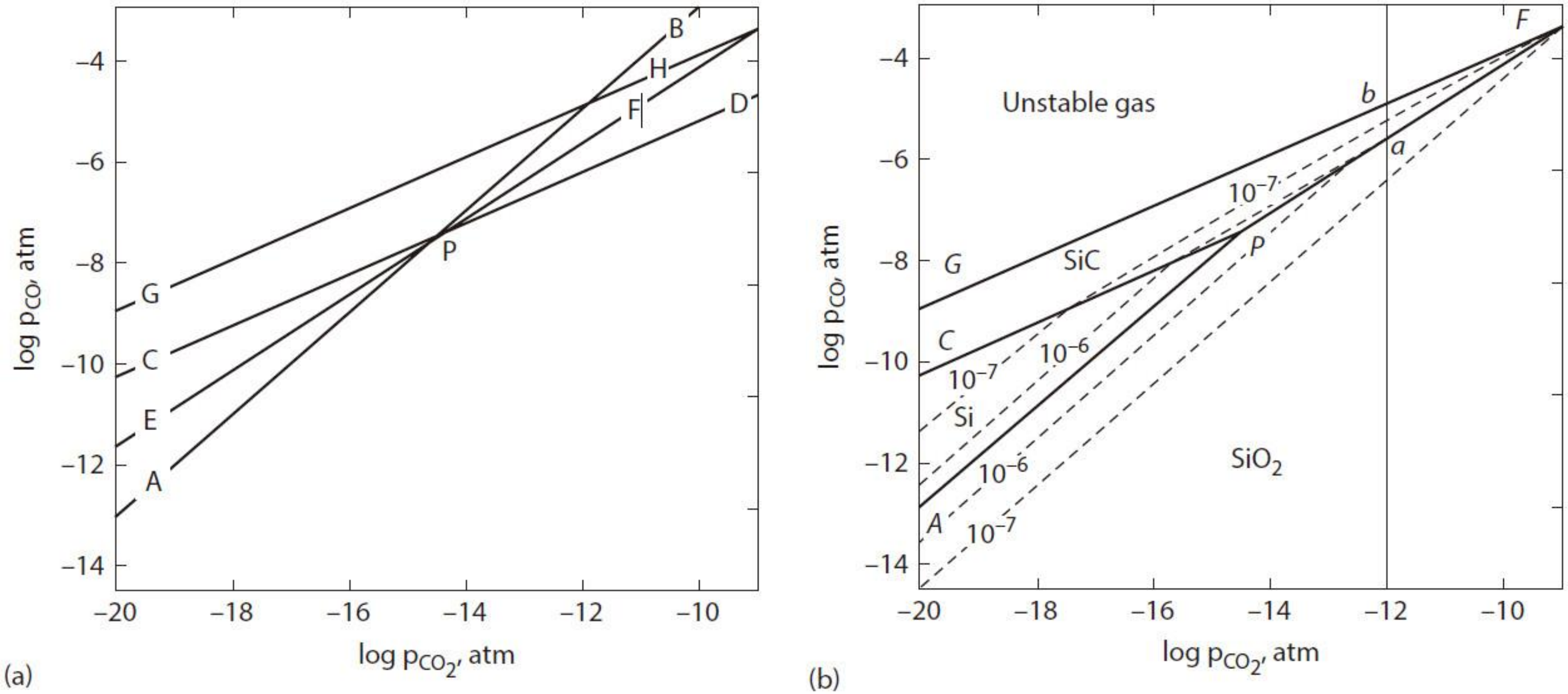


**Figure 13.6** (a) Construction of the phase stability diagram for the system Si–C–O at 1273 K. (b) The phase stability diagram for the system Si–C–O at 1273 K showing the $10^{-6}$ and $10^{-7}$ atm SiO isobars.

# 13.5 Phase Stability Diagrams

- ✓ The equivalent of the equilibrium given by Equation (ii) is

$$\mathrm{Si}_{(s)} + 2\mathrm{CO}_{(g)} = \mathrm{SiC}_{(s)} + \mathrm{CO}_{2(g)} \qquad \text{(x)}$$

for which

$$\Delta G^{\circ}_{(\mathrm{x})1273\ \mathrm{K}} = -11,900\ \mathrm{J} = -8.3144 \times 1273 \times 2.303 \log \frac{p_{\mathrm{CO}_2}}{p^2_{\mathrm{CO}}}$$

or

$$\log p_{\mathrm{CO}} = \frac{1}{2} \log p_{\mathrm{CO}_2} - 0.24$$

- ✓ This is drawn as line *CD* in Figure 13.6a. Above the line, SiC is stable relative to Si, and below the line, Si is stable relative to SiC. The lines *AB* and *CD* intersect at *P*, which is thus the invariant point at which Si, SiC, $SiO_2$, and a gas phase are in equilibrium.

# 13.5 Phase Stability Diagrams

✓ The equivalent of the equilibrium given by Equation (iii) is

$$SiC_{(s)} + 3CO_{2(g)} = SiO_{2(s)} + 4CO_{(g)} \qquad \text{(xi)}$$

for which

$$\Delta G^\circ_{(\text{xi})1273\text{ K}} = -327,700\text{ J} = -8.3144 \times 1273 \times 2.303 \log \frac{p^4_{CO}}{p^3_{CO_2}}$$

or

$$\log p_{CO} = 0.75 \log p_{CO_2} + 3.36$$

✓ This is drawn as the line *EF* in Figure 13.6a. SiC is stable relative to $SiO_2$ above the line, and $SiO_2$ is stable relative to SiC below the line.

# 13.5 Phase Stability Diagrams

✓ The *carbon line*, which is equivalent to the $\log a_{\mathrm{C}} = 0$ line in Figure 13.5, is obtained from

$$\mathrm{C}_{(s)} + \mathrm{CO}_{2(g)} = 2\mathrm{CO}_{(g)} \qquad \text{(xii)}$$

for which

$$\Delta G^{\circ}_{(\mathrm{xii})1273\ \mathrm{K}} = -51,400\ \mathrm{J} = -8.3144 \times 1273 \times 2.303 \log \frac{p^2_{\mathrm{CO}}}{p_{\mathrm{CO}_2}}$$

which gives

$$\log p_{\mathrm{CO}} = \frac{1}{2} \log p_{\mathrm{CO}_2} - 1.05$$

✓ This is drawn as the line *GH* in Figure 13.6a.

# 13.5 Phase Stability Diagrams

- ✓ The phase stability fields, identified in the same manner as was used in Figure 13.5a, are shown in Figure 13.6b. The Si phase field is bounded by *APC*, the SiC phase field is bounded by *CPFG*, and the $SiO_2$ phase field lies below the line *APF*. The region above *GF* is an unstable gas. Any gas in this region precipitates carbon according to

$$2CO_{(g)} \rightarrow CO_{2(g)} + C_{(s)}$$

until, thereby, the ratio $p_{CO}/p_{CO_2}$ is that required for equilibrium with C at $a_C = 1$ at 1273 K.

- ✓ The isobars for SiO gas are calculated as follows.

# 13.5 Phase Stability Diagrams

✓ In the field of stability of Si, the equivalent of the equilibrium given by Equation (vi) is

$$\mathrm{Si}_{(s)} + \mathrm{CO}_{2(g)} = \mathrm{SiO}_{(g)} + \mathrm{CO}_{(g)} \qquad \text{(xiii)}$$

for which

$$\Delta G^{\circ}_{(\text{xiii})1273\ \mathrm{K}} = -37,300\ \mathrm{J} = -8.3144 \times 1273 \times 2.303 \log \frac{p_{\mathrm{SiO}}\, p_{\mathrm{CO}}}{p_{\mathrm{CO_2}}}$$

which gives

$$\log p_{\mathrm{CO}} = \log p_{\mathrm{CO_2}} + 1.53 - \log p_{\mathrm{SiO}}$$

# 13.5 Phase Stability Diagrams

✓ In the field of stability of $SiO_2$, the equivalent of the equilibrium given by Equation (vii) is

$$SiO_{2(s)} + CO_{(g)} = SiO_{(g)} + CO_{2(g)} \qquad \text{(xiv)}$$

for which

$$\Delta G^{\circ}_{(\mathrm{xiv})1273\ \mathrm{K}} = 302,300\ \mathrm{J} = -8.3144 \times 1273 \times 2.303 \log \frac{p_{\mathrm{CO_2}}\, p_{\mathrm{SiO}}}{p_{\mathrm{CO}}}$$

which gives

$$\log p_{\mathrm{CO}} = \log p_{\mathrm{CO_2}} + 12.40 + \log p_{\mathrm{SiO}}$$

✓ In the field of stability of SiC, the equilibrium equivalent to that given by Equation (viii) is

$$SiC_{(s)} + 2CO_{2(g)} = SiO_{(g)} + 3CO_{(g)} \qquad \text{(xv)}$$

for which

$$\Delta G^{\circ}_{(xv)1273\,K} = -25,400\ \text{J} = -8.3144 \times 1273 \times 2.303 \log \frac{p^3_{CO}\, p_{SiO}}{p^2_{CO_2}}$$

which gives

$$\log p_{CO} = \frac{2}{3} \log p_{CO_2} - 0.35 + \frac{1}{3} \log p_{SiO}$$

# 13.5 Phase Stability Diagrams

- ✓ The $10^{-7}$ and $10^{-6}$ atm SiO isobars are shown in Figure 13.6b, and a comparison of Figures 13.5 and 13.6 shows that the latter is produced by distorting the former such that the CO isobars are horizontal lines and the $CO_2$ isobars are vertical lines. The points *P* and *F* in Figure 13.6b correspond to the points *P* and *F* in Figure 13.5.

- ✓ The full phase stability diagram is three-dimensional, with the axes being temperature and the activities of two components ( $a_C$ and $p_{O_2}$) or the partial pressures of CO and $CO_2$.

- ✓ Two-dimensional phase stability diagrams which show the influence of temperature can be drawn if the activity of one component is held constant.

# 13.5 Phase Stability Diagrams

- ✓ Consider the phase stability diagram using $p_{\mathrm{CO}}$ and 1/*T* as coordinates and drawn for a constant partial pressure of $CO_2$ of $10^{-12}$ atm.

- ✓ The diagram will be drawn for temperatures in the range 1250–1990 K, and thus, since the melting temperature of Si is 1683 K, equilibria involving both solid and liquid Si will have to be considered.

- ✓ In Figure 13.7a, the vertical line at $T$ = 1685 K represents the melting temperature of Si, and thus, liquid Si occurs to the left of the line and solid Si occurs to the right.

**1. The equilibrium $Si_{(s)}$–$SiO_2$–gas phase**

The equilibrium is

$$Si_{(s)} + 2CO_{2(g)} = SiO_{2(s)} + 2CO_{(g)} \qquad \text{(xvi)}$$

for which

$$\Delta G^{\circ}_{(\text{xvi})} = -337{,}300 - 0.02T \text{ J}$$

Thus,

$$\frac{-337{,}300}{8.3144 \times 2.303T} - \frac{0.02}{8.3144 \times 2.303} = -\log \frac{p_{CO}^2}{p_{CO_2}^2}$$

which can be rearranged to give

$$\log p_{CO} = \frac{8807}{T} + 5.22 \times 10^{-4} + \log p_{CO_2}$$

This line, with $\log p_{CO_2} = -12$, is drawn as *AB* in Figure 13.7a. Silicon is stable relative to $SiO_2$ above the line, and $SiO_2$ is stable relative to Si below the line.

# 13.5 Phase Stability Diagrams

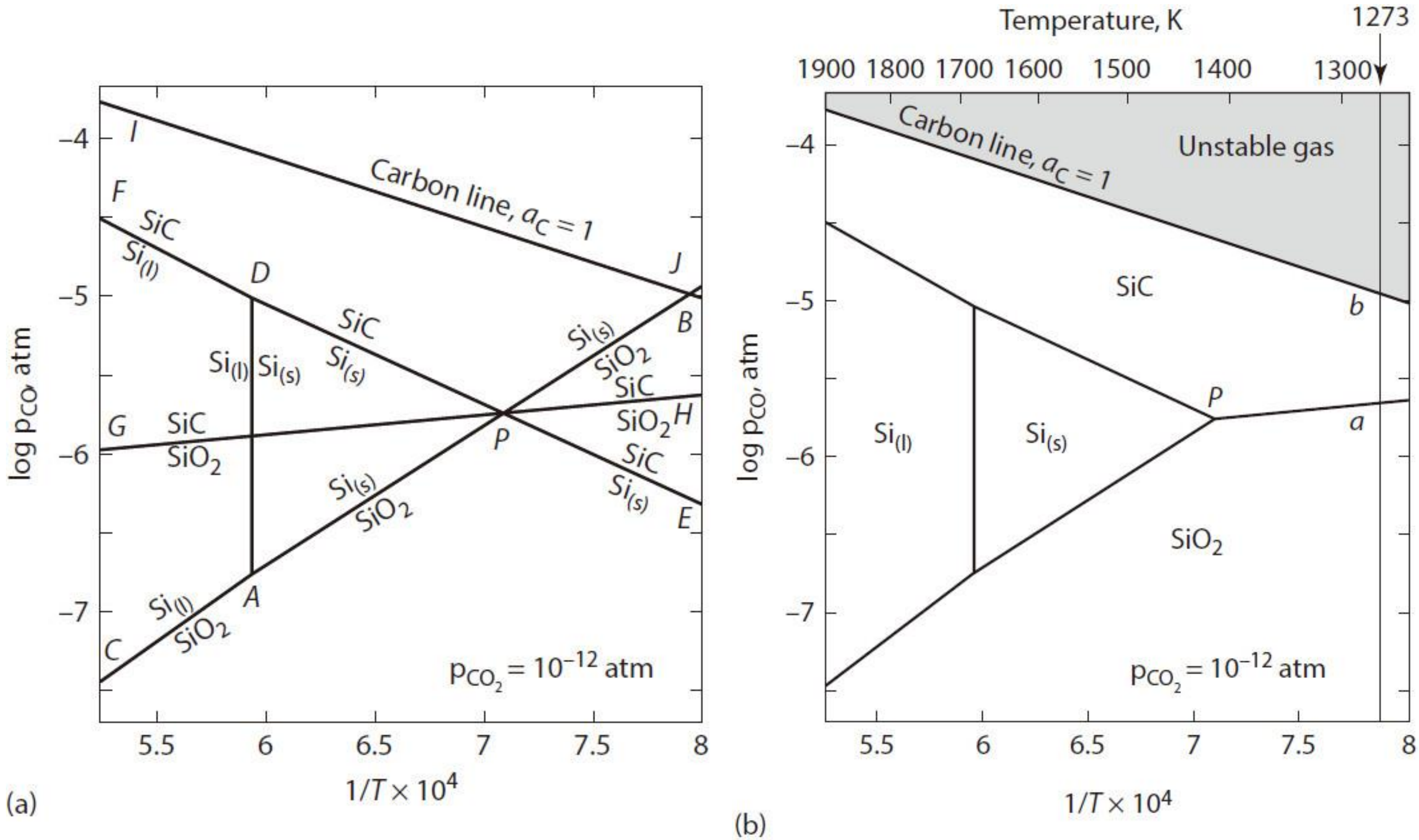


**Figure 13.7** (a) Construction of the phase stability diagram for the system Si–C–O using log $p_{CO}$ and $1/T$ as coordinates at a constant partial pressure of $CO_2$ of $10^{-12}$ atm. (b) The phase stability diagram for the system Si–C–O at a constant partial pressure of $CO_2 = 10^{-12}$ atm.

# 13.5 Phase Stability Diagrams

✓ The equilibrium $Si_{(l)}$ –$SiO_2$ –gas phase: The equilibrium is

$$\mathrm{Si}_{(l)} + 2\mathrm{CO}_{2(g)} = \mathrm{SiO}_{2(s)} + 2\mathrm{CO}_{(g)} \qquad \text{(xvii)}$$

for which

$$\Delta G^{\circ}_{(\mathrm{xvii})} = -387,900 + 30.18T \text{ J}$$

Thus,

$$\frac{-387,900}{8.3144 \times 2.303T} + \frac{30.18}{8.3144 \times 2.303} = -\log \frac{p^2_{\mathrm{CO}}}{p^2_{\mathrm{CO_2}}}$$

which gives

$$\log p_{\mathrm{CO}} = \frac{10,130}{T} - 0.788 + \log p_{\mathrm{CO_2}}$$

With $p_{\mathrm{CO}} = 10^{-12}$ atm, this gives line *CA* in Figure 13.7a. Liquid Si is stable relative to $SiO_2$ above the line, and $SiO_2$ is stable relative to liquid Si below the line.

**2. The equilibrium $Si_{(s)}$–SiC–gas phase**

The equilibrium is

$$Si_{(s)} + 2CO_{(g)} = SiC_{(s)} + CO_{2(g)} \qquad \text{(xviii)}$$

for which

$$\Delta G^{\circ}_{(\mathrm{xviii})} = -243{,}750 + 182.11T \text{ J}$$

which yields

$$\log p_{CO} = -\frac{6364}{T} + 4.76 + \frac{1}{2}\log p_{CO_2}$$

With $\log p_{CO_2} = -12$ , this gives line *DE* in Figure 13.7a. SiC is stable relative to solid Si above the line, and Si is stable relative to SiC below the line.

**3. The equilibrium $Si_{(l)}$–SiC–gas phase:**

The equilibrium is

$$Si_{(l)} + 2CO_{(g)} = SiC_{(s)} + CO_{2(g)} \qquad \text{(xix)}$$

for which

$$\Delta G^{\circ}_{(\text{xix})} = -293{,}300 + 211.5T \text{ J}$$

This gives

$$\log p_{CO} = -\frac{7659}{T} + 5.52 + \frac{1}{2}\log p_{CO_2}$$

which, with $\log p_{CO_2} = -12$, gives line *FD* in Figure 13.7a. Liquid Si is stable with respect to SiC below the line, and SiC is stable with respect to Si above the line.

**4. The equilibrium SiC–$SiO_2$–gas phase:**

The equilibrium is

$$SiC_{(s)} + 3CO_{2(g)} = SiO_{2(s)} + 4CO_{(g)} \qquad \text{(xx)}$$

which gives

$$\log p_{CO} = \frac{1221}{T} + 2.38 + 0.75 \log p_{CO_2}$$

This is drawn, with $\log p_{CO_2} = -12$, as line *GH* in Figure 13.7a. SiC is stable relative to $SiO_2$ above the line, and $SiO_2$ is stable relative to SiC below the line.

# 13.5 Phase Stability Diagrams

✓ The carbon line, at which $a_{\mathrm{C}} = 1$, is determined by the equilibrium

$$\mathrm{C}_{(s)} + \mathrm{CO}_{2(g)} = 2\mathrm{CO}_{(g)}$$

for which

$$\Delta G^{\circ} = 170,700 - 174.5T \ \mathrm{J}$$

which gives

$$\log p_{\mathrm{CO}} = -\frac{4457}{T} + 4.55 + \frac{1}{2}\log p_{\mathrm{CO}_2}$$

With $p_{\mathrm{CO}} = 10^{-12}$ atm, this gives line *IJ* in Figure 13.7a. States above this line represent unstable gas.

# 13.5 Phase Stability Diagrams

✓ The stability fields are identified as follows:

1. In the area *APD*, solid silicon is stable with respect to $SiO_2$ and SiC, and thus, this area is the field of stability of solid Si.
2. In the areas *FDAC*, liquid silicon is stable with respect to SiC and $SiO_2$, and thus, this area is the field of stability of liquid Si.
3. In the area *FDPHJI*, SiC is stable with respect to Si and $SiO_2$, and thus, this area is the field of stability of SiC.
4. Below the line *CAPH*, $SiO_2$ is stable with respect to Si and SiC, and thus, this is the field of stability of $SiO_2$.

# 13.5 Phase Stability Diagrams

- ✓ These fields of stability are shown in Figure 13.7b. The phase stability diagram shows that, with $p_{\mathrm{CO}} = 10^{-12}$ atm, 1410 K (the point *P*) is the minimum temperature at which silicon is stable, and that, with increasing temperature, the field of stability of silicon widens at the expense of SiC and $SiO_2$.

- ✓ The slopes of the lines in Figures 13.7 are related to the standard enthalpy changes for the equilibrium reactions; for example, the slope of the line dividing the solid Si and the $SiO_2$ fields (the line *AB* in Figure 13.7a) is obtained as $-\left(\Delta H^0/\left(2.303\cdot 2\cdot R\right)\right) = 337,300/\left(2.303\times 2\times R\right) = 8807$ for the reaction given by Equation (xvi).

- ✓ The points *a* and *b* on the $10^{-12}$ atm $CO_2$ isobar in Figure 13.6b correspond with the points *a* and *b* on the 1273 K isotherm in Figure 13.7b.

# Outlines

# 13.6 Binary Systems Containing Compounds

- ✓ The phase relationships in a two-component system can be represented on an isobaric phase diagram using temperature and composition as coordinates, and these are the phase diagrams often encountered in materials science (see Chapter 10).
- ✓ If the two components react with one another to form compounds, then, in such systems, chemical reaction equilibria and phase equilibria are synonymous.
- ✓ Consider the binary system $A – B$, the phase diagram for which is shown in Figure 13.8. The negative departures from ideality in the solid state are sufficiently large that stoichiometric compounds following the law of definite proportions are formed, with there being negligible solubility of *A* in *B*, or *B* in *A*, and negligible range of nonstoichiometry in the compounds $AB_3$, $AB$, and $A_3B$.

# 13.6 Binary Systems Containing Compounds

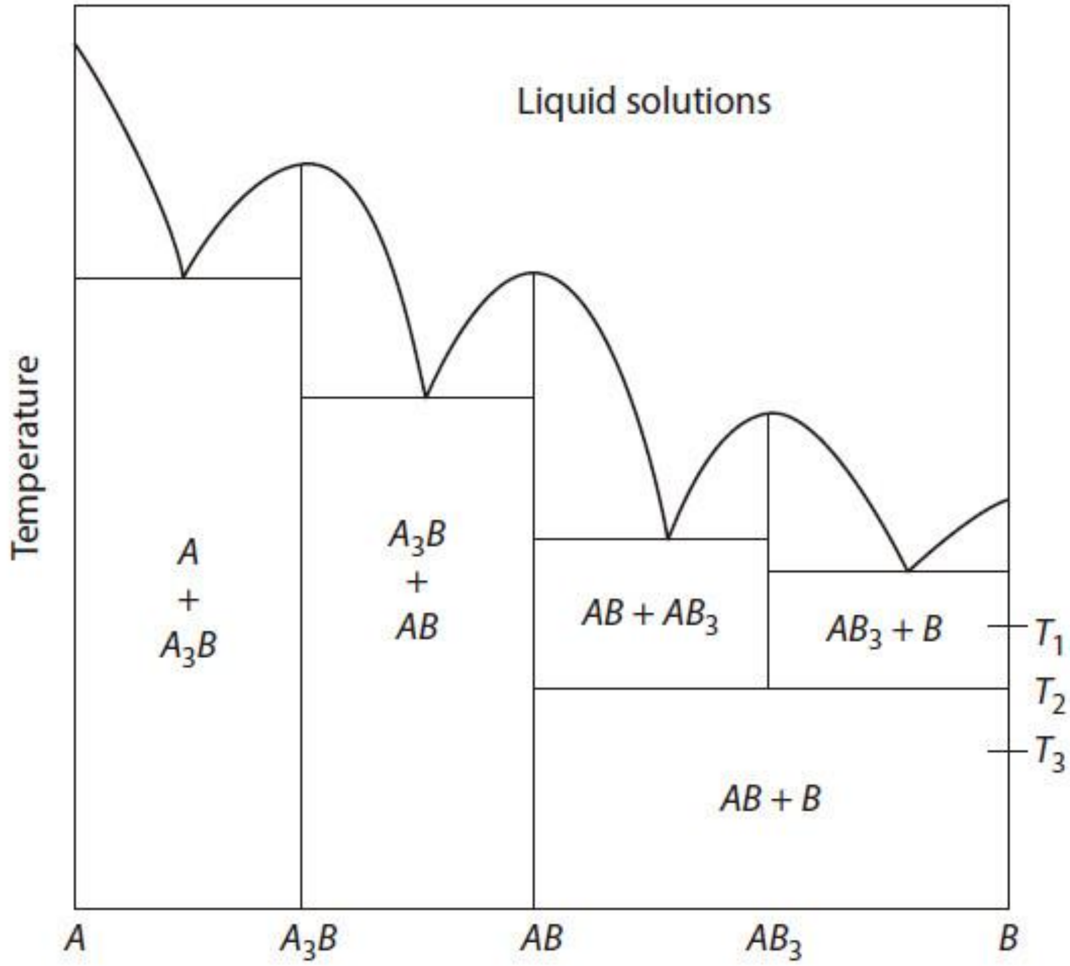


**Figure 13.8** The phase diagram for the system *A*–*B* in which three stoichiometric compounds are formed.

# 13.6 Binary Systems Containing Compounds

- The system contains the equilibria

$$3A + B = A_3B$$

$$A + B = AB$$

$$A + 3B = AB_3$$

- If one of the components (*B*) is appreciably volatile and the other (*A*) is not, then the thermodynamics of the system can be determined from knowledge of the variation of $p_B$ with composition.
- The variation of $p_B$ with composition at the temperature $T_1$ is shown in Figure 13.9. In the range of composition between *B* and $AB_3$, virtually pure *B* exists in equilibrium with $AB_3$ (saturated with *B*), and, since pure *B* exists, the pressure exerted by the system is $p_B$, the saturated vapor pressure of *B* at the temperature $T_1$.

# 13.6 Binary Systems Containing Compounds

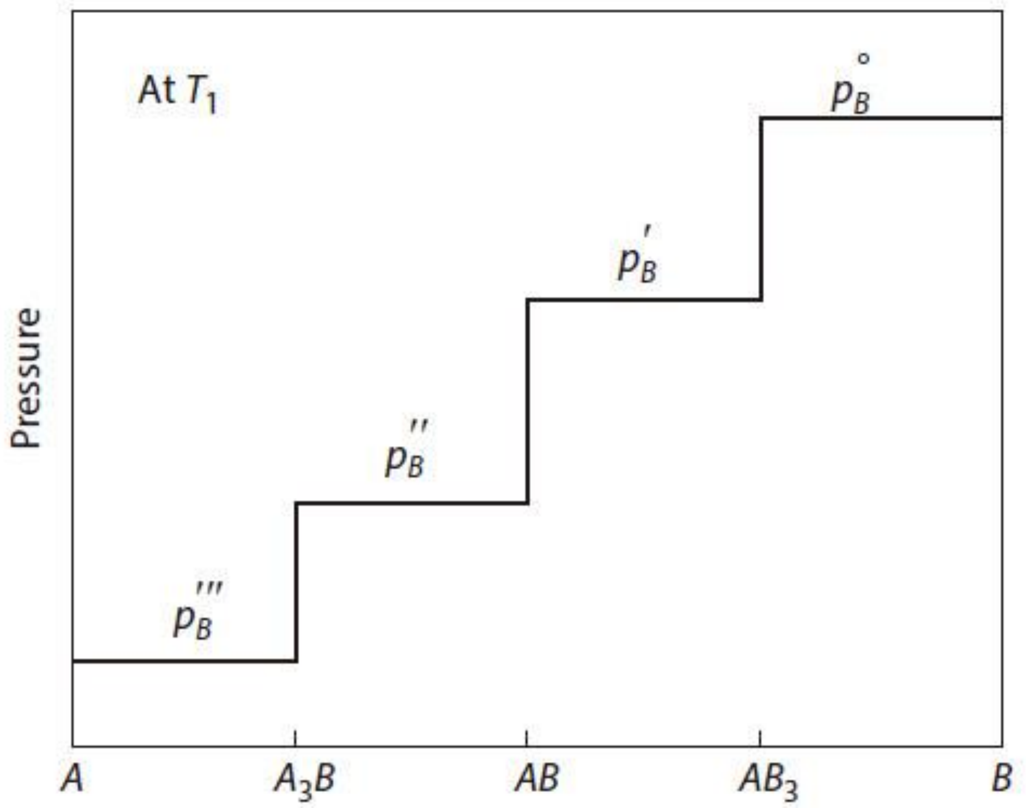


**Figure 13.9** The variation, with composition, of the vapor pressure of component $B$ in the system shown in Figure 13.8 at the temperature $T_1$.

# 13.6 Binary Systems Containing Compounds

- ✓ In the range of composition between $AB$ (saturated with $B$) and $AB_3$ (saturated with $A$), the constant pressure exerted by the system is $p'_B$, in the range between $A_3B$ (saturated with $B$) and $AB$ (saturated with $A$) it is $p''_B$, and in the range between $A$ (saturated with $B$) and $A_3B$ (saturated with $A$) it is $p'''_B$.
- ✓ In each of these ranges of composition, the two-component, three-phase equilibrium has one degree of freedom, which is used when $T_1$ is specified, which requires that $P = p_B$ within these ranges at fixed temperature.

# 13.6 Binary Systems Containing Compounds

✓ The activity of *B* in the system, defined as $p_B / p_B^\circ$ , is thus

$$\frac{p_B}{p_B^\circ} = 1 \text{ in the range } AB_3 - B$$

$$\frac{p'_B}{p_B^\circ} \text{ in the range } AB - AB_3$$

$$\frac{p''_B}{p_B^\circ} \text{ in the range } A_3B - AB$$

$$\frac{p'''_B}{p_B^\circ} \text{ in the range } A - A_3B$$

# 13.6 Binary Systems Containing Compounds

✓ Thus, since $\Delta \overline{G}_B^M = \overline{G}_B - G_B^\circ = RT \ln p_B / p_B^\circ$, the Gibbs free energy–composition diagram at the temperature $T_1$ is as shown in Figure 13.10. Since Figure 13.10 is drawn for 1 mole of the system, then

$$hb = \Delta G_{(\mathrm{i})} \quad = \Delta G \text{ for the reaction}$$

$$0.75 \text{ mole } A + 0.25 \text{ mole } B = A_{0.75}B_{0.25} \text{ or } 0.25 \text{ moles of } A_3B$$

$$gc = \Delta G_{(\mathrm{ii})} \quad = \Delta G \text{ for the reaction}$$

$$0.5 \text{ mole } A + 0.5 \text{ mole } B = A_{0.5}B_{0.5} \text{ or } 0.5 \text{ moles of } AB$$

$$fd = \Delta G_{(\mathrm{iii})} \quad = \Delta G \text{ for the reaction}$$

$$0.25 \text{ mole } A + 0.75 \text{ mole } B = A_{0.25}B_{0.75} \text{ or } 0.25 \text{ moles of } AB_3$$

# 13.6 Binary Systems Containing Compounds

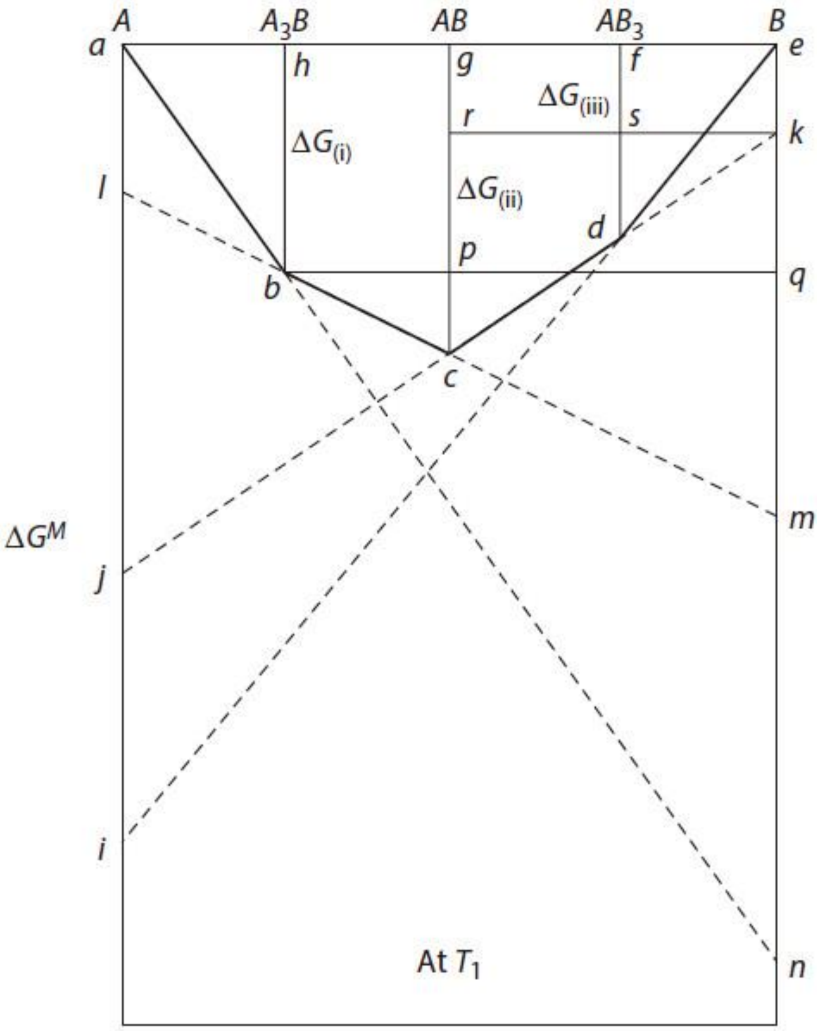


**Figure 13.10** The molar Gibbs free energies at $T_1$ in the system shown in Figure 13.8.

# 13.6 Binary Systems Containing Compounds

- ✓ These three changes in Gibbs free energy can be determined geometrically as follows:

$$ek = \Delta \bar{G}_B'^{\,M} = RT \ln \frac{p_B'}{p_B^\circ}$$

$$em = \Delta \bar{G}_B''^{\,M} = RT \ln \frac{p_B''}{p_B^\circ}$$

and

$$en = \Delta \bar{G}_B'''^{\,M} = RT \ln \frac{p_B'''}{p_B^\circ}$$

- ✓ Thus, from consideration of the similar triangles *ahb* and *aen*,

$$\frac{\Delta G_{(\mathrm{i})}}{\Delta \bar{G}_B'''^{\,M}} = \frac{1}{4}$$

and hence,

$$\Delta G_{(\mathrm{i})} = \frac{1}{4} \Delta \bar{G}_B'''^{\,M}$$

# 13.6 Binary Systems Containing Compounds

- Consideration of the similar triangles *bpc* and *bqm* gives

$$\frac{pc}{qm} = \frac{bp}{bq} = \frac{1}{3}$$

- But

$$pc = gc - gp = \Delta G_{(\mathrm{ii})} - \Delta G_{(\mathrm{i})}$$

and

$$qm = em - eq = \Delta \bar{G}''^{\,M}_{B} - \Delta G_{(\mathrm{i})}$$

- Thus,

$$3\left(\Delta G_{(\mathrm{ii})} - \Delta G_{(\mathrm{i})}\right) = \left(\Delta \bar{G}''^{\,M}_{B} - \Delta G_{(\mathrm{i})}\right)$$

or

$$\Delta G_{(\mathrm{ii})} = \frac{1}{3}\Delta \bar{G}''^{\,M}_{B} + \frac{2}{3}\Delta G_{(\mathrm{i})}$$

# 13.6 Binary Systems Containing Compounds

- Consideration of the similar triangles *rck* and *sdk* gives

$$\frac{rc}{sd} = \frac{rk}{sk} = 2$$

- But

$$rc = gc - gr = \Delta G_{(\mathrm{ii})} - \Delta \bar{G}_B'^{M}$$

and

$$sd = fd - fs = \Delta G_{(\mathrm{iii})} - \Delta \bar{G}_B'^{M}$$

such that

$$\Delta G_{(\mathrm{iii})} = \frac{1}{2}\Delta G_{(\mathrm{ii})} + \frac{1}{2}\Delta \bar{G}_B'^{M}$$

# 13.6 Binary Systems Containing Compounds

✓ Thus,

$$\Delta G_{(\mathrm{i})} = \frac{1}{4} RT \ln \frac{p_B'''}{p_B^\circ}$$

$$\Delta G_{(\mathrm{ii})} = \frac{1}{3} RT \ln \frac{p_B''}{p_B^\circ} + \frac{1}{6} RT \ln \frac{p_B'''}{p_B^\circ}$$

and

$$\Delta G_{(\mathrm{iii})} = \frac{1}{6} RT \ln \frac{p_B''}{p_B^\circ} + \frac{1}{12} RT \ln \frac{p_B'''}{p_B^\circ} + \frac{1}{2} RT \ln \frac{p_B'}{p_B^\circ}$$

and hence for

$$3A + B = A_3B \quad \Delta G^\circ_{(\mathrm{i})} = 4\Delta G_{(\mathrm{i})}$$

$$A + B = AB \quad \Delta G^\circ_{(\mathrm{ii})} = 2\Delta G_{(\mathrm{ii})} = \frac{2}{3}\Delta \bar{G}_B''^{\,M} + \frac{4}{3}\Delta G_{(\mathrm{i})}$$

$$A + 3B = AB_3 \quad \Delta G^\circ_{(\mathrm{iii})} = 4\Delta G_{(\mathrm{iii})} = 2\Delta G_{(\mathrm{ii})} + 2\Delta \bar{G}_B'^{\,M}$$

# 13.6 Binary Systems Containing Compounds

- ✓ Figure 13.8 shows that, below the temperature $T_2$, the compound $AB_3$ is unstable with respect to $AB$ and $B$, with the invariant equilibrium

$$AB_{(s)} + 2B_{(s)} = AB_{3(s)}$$

occurring at the temperature $T_2$. The standard Gibbs free energy change for this reaction is calculated as $\Delta G^\circ_{\text{(iii)}} - \Delta G^\circ_{\text{(ii)}}$, which equals $2RT\ln p'_B / p^\circ_B$ .

- ✓ Thus, at $T > T_2$,

$$\Delta G^\circ_{\text{(iii)}} - \Delta G^\circ_{\text{(ii)}} < 0 \quad \text{and} \quad p'_B < p^\circ_B$$

at $T_2$,

$$\Delta G^\circ_{\text{(iii)}} - \Delta G^\circ_{\text{(ii)}} = 0 \quad \text{and} \quad p'_B = p^\circ_B$$

and at T < $T_2$,

$$\Delta G^\circ_{\text{(iii)}} - \Delta G^\circ_{\text{(ii)}} > 0 \quad \text{and} \quad p'_B > p^\circ_B$$

# 13.6 Binary Systems Containing Compounds

- ✓ The variations of the Gibbs free energy of mixing with temperature at $T_2$ and $T_3$ are shown in Figure 13.11a and b, respectively, which illustrates graphically that at $T_2$,

$$\Delta G_{(iii)} = \tfrac{1}{2}\Delta G_{(ii)}$$

and at $T_3$,

$$\left|\Delta G_{(iii)}\right| < \left|\tfrac{1}{2}\Delta G_{(ii)}\right|$$

such that, in the range of composition $B - AB$, at temperatures below $T_2$, the system occurring as either $AB + AB_3$ or $AB_3 + B$ is metastable with respect to its occurrence as $AB + B$.

- ✓ In considering the thermodynamic properties of a system such as that shown in Figure 13.8, two approaches can be made: namely,

1. The consideration that the compounds are ordered solid solutions
2. The consideration that the compounds are formed by the chemical reaction of $A$ with $B$

# 13.6 Binary Systems Containing Compounds

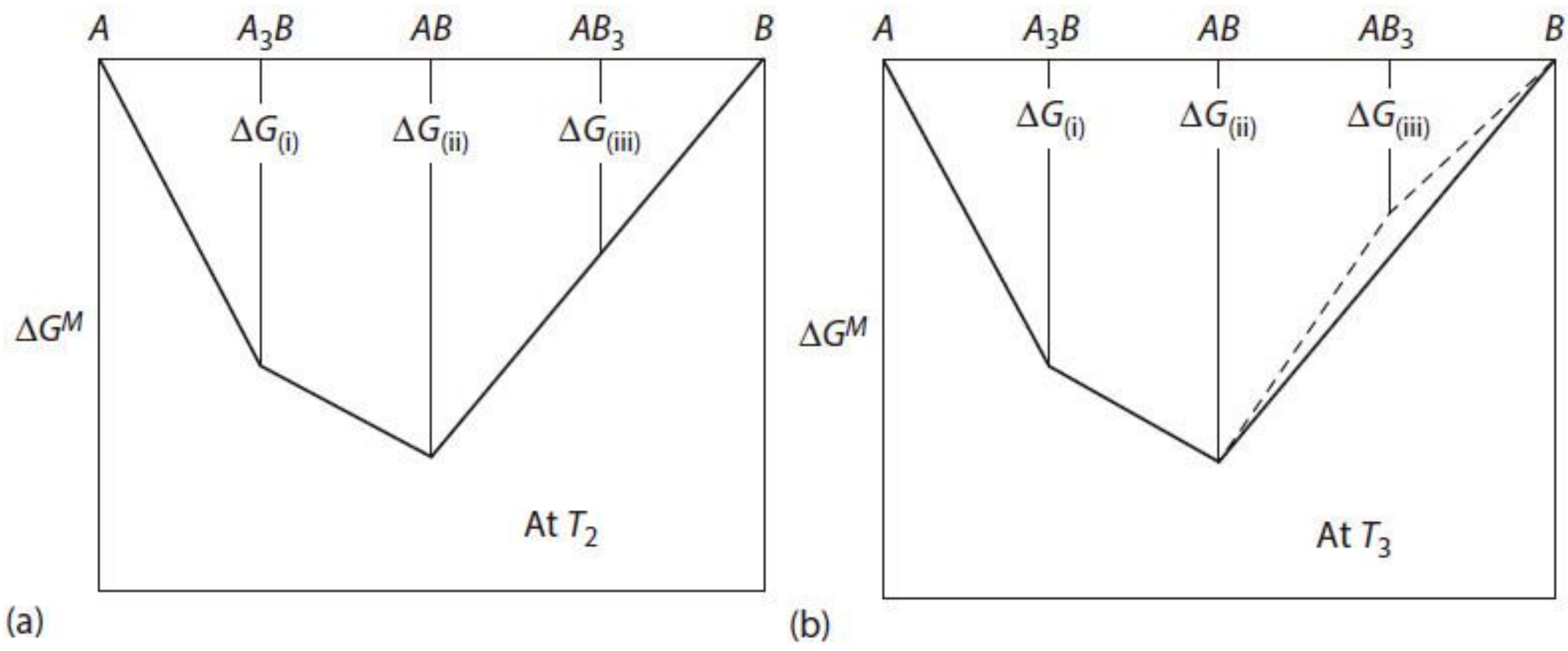


**Figure 13.11** (a) The Gibbs free energies of mixing in the system shown in Figure 13.8 at the temperature $T_2$. (b) The Gibbs free energies of mixing in the system shown in Figure 13.8 at the temperature $T_3$.

# 13.6 Binary Systems Containing Compounds

1. Consider the compound $AB_3$ to be an ordered solid solution of *A* and *B* in the molar ratio 1/3. Then, in Figure 13.10,

$$fd = \Delta G^M = RT\left(X_A \ln a_A + X_B \ln a_B\right) = RT\left(0.25 \ln a_A + 0.75 \ln a_B\right) = RT \ln a_A^{0.25} a_B^{0.75} \qquad \text{(i)}$$

2. Consider the compound $AB_3$ to form as the product of the reaction

$$A + 3B = AB_3$$

for which the change in the standard Gibbs free energy is $\Delta G^\circ_{(\text{iii})}$. Then,

$$fd = 0.25\Delta G^\circ_{(\text{iii})} = -0.25RT \ln K_{(\text{iii})} = -RT \ln\left(\frac{a_{AB_3}}{a_A a_B^3}\right)^{0.25} = RT \ln\left(\frac{a_A^{0.25} a_B^{0.75}}{a_{AB_3}^{0.25}}\right) \qquad \text{(ii)}$$

# 13.6 Binary Systems Containing Compounds

- Since $AB_3$ is a *line compound* (i.e., it has a negligible range of nonstoichiometry), it exists at a fixed composition and hence exists in a fixed state.
- If this fixed state is chosen as being the standard state, in which $a_{AB_3} = 1$, then Equation (ii) becomes

$$fd = 0.25\Delta G^{\circ}_{(\text{iii})} = RT \ln a_A^{0.25} a_B^{0.75}$$

which is identical with Equation (i).

- In both Equations (i) and (ii), the standard states of *A* and *B* are the pure solid elements at the temperature *T*.
- The variations of the activities of *A* and *B* in the compound $AB_3$ are limited by the separation of *B* and *AB*; for example, when $AB_3$ is in equilibrium with *B*, $a_B = 1$, and thus, $a_A \exp\left(4\Delta G_{(\text{iii})}/RT\right) = \exp\left(4\Delta G^{\circ}_{(\text{iii})}/RT\right)$ ($RT \ln a_A = ai$ in Figure 13.10).

# 13.6 Binary Systems Containing Compounds

- ✓ If the activity of *B* is decreased to a value less than unity, then $AB_3$ is no longer saturated with *B*, and the activity of *A* in the compound increases in accordance with Equation (iv).
- ✓ The minimum activity of *B* in $AB_3$ is determined by the saturation of $AB_3$ with *A*, at which point the compound *AB* appears. The minimum activity which *B* may have is obtained from Figure 13.10 as $RT \ln a_B = ek$, and the corresponding maximum activity of *A* is obtained from $RT \ln a_A = aj$.
- ✓ Nonsaturation of $AB_3$ with either *A* or *B* occurs when the partial pressure of *B* exerted by the compound lies between the limits $p'_B$ and $p^{\circ}_B$ .

# 13.6 Binary Systems Containing Compounds

- ✓ Similar consideration can be made with respect to the compounds *AB* and $A_3B$; for example, in Figure 13.10,

$$gc = \Delta G^M = \frac{1}{2}\Delta G^\circ_{(ii)} = RT \ln a_A^{0.5} a_B^{0.5}$$

and

$$hb = \Delta G^M = 0.25\Delta G^\circ_{(i)} = RT \ln a_A^{0.75} a_B^{0.25}$$

- ✓ In the preceding discussion, it was assumed that the intermediate phases were line compounds and that there was no solubility of *A* in *B*. If *A* and *B* are partially soluble in one another and the compounds $A_2B$ and $AB_2$ (identified as phases β and γ, respectively) have measurable ranges of nonstoichiometry, the phase diagram is as shown in Figure 13.12a.
- ✓ Again, if *B* is appreciably volatile and *A* is not, then the variation of vapor pressure with composition at the temperature $T_1$ is as shown in Figure 13.12b, and the corresponding variation of the Gibbs free energy is as shown in Figure 13.12c.

# 13.6 Binary Systems Containing Compounds

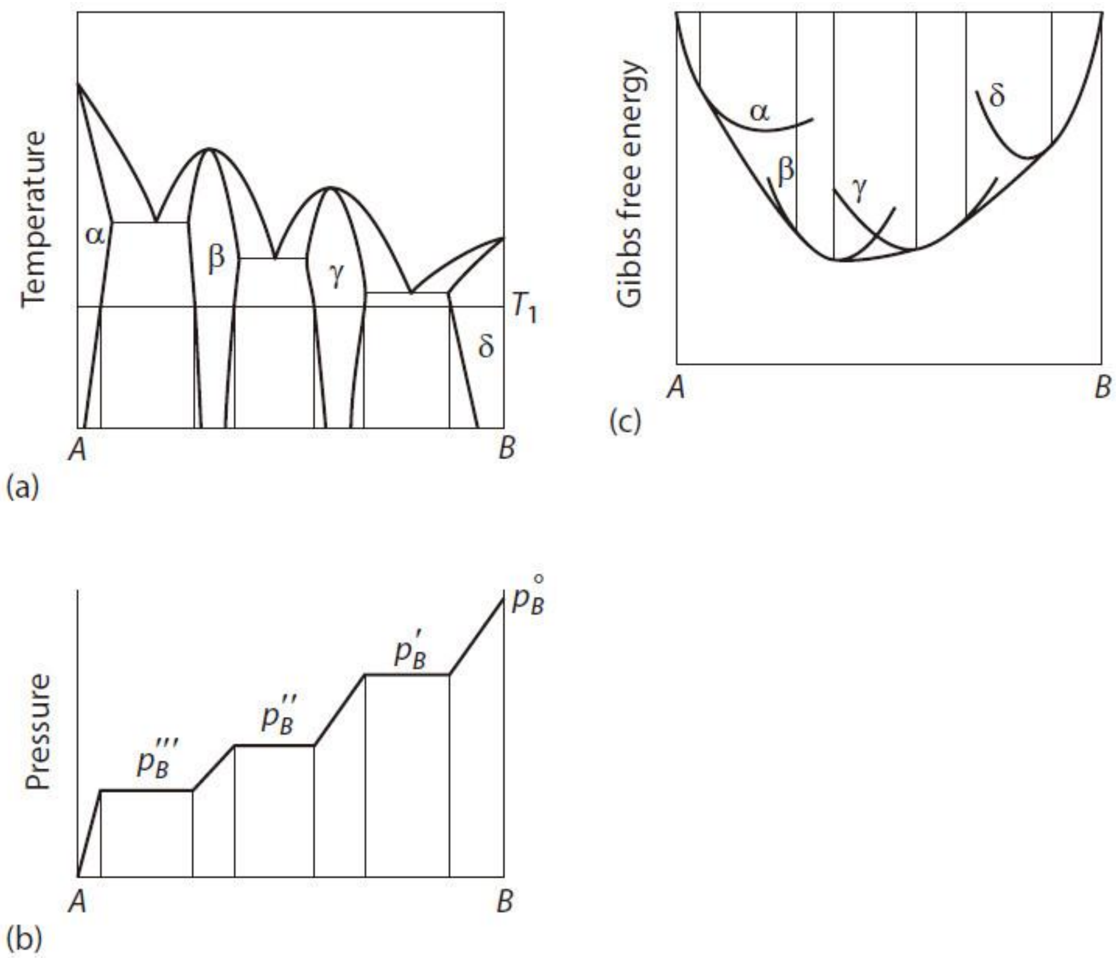


**Figure 13.12** (a) The phase diagram for the system *A*–*B*. (b) The partial vapor pressure of *B* at the temperature $T_1$. (c) The molar Gibbs free energies of mixing at the temperature $T_1$.

**Example 1: The Ga–GaP System**

The phase diagram for the system Ga–GaP is shown in Figure 13.13. Calculate the partial pressure of phosphorus vapor, $p_{P_2}$, exerted by the GaP liquidus melt at 1273 K. The standard Gibbs free energy change for the reaction

$$\mathrm{Ga}_{(l)} + \frac{1}{2}\mathrm{P}_{2(g)} = \mathrm{GaP}_{(s)}$$

is

$$\Delta G^\circ = -178{,}800 + 96.2T + 3.1T\ln T - 3.61\times10^{-3}T^2 - \frac{1.035\times10^5}{T}$$

# 13.6 Binary Systems Containing Compounds

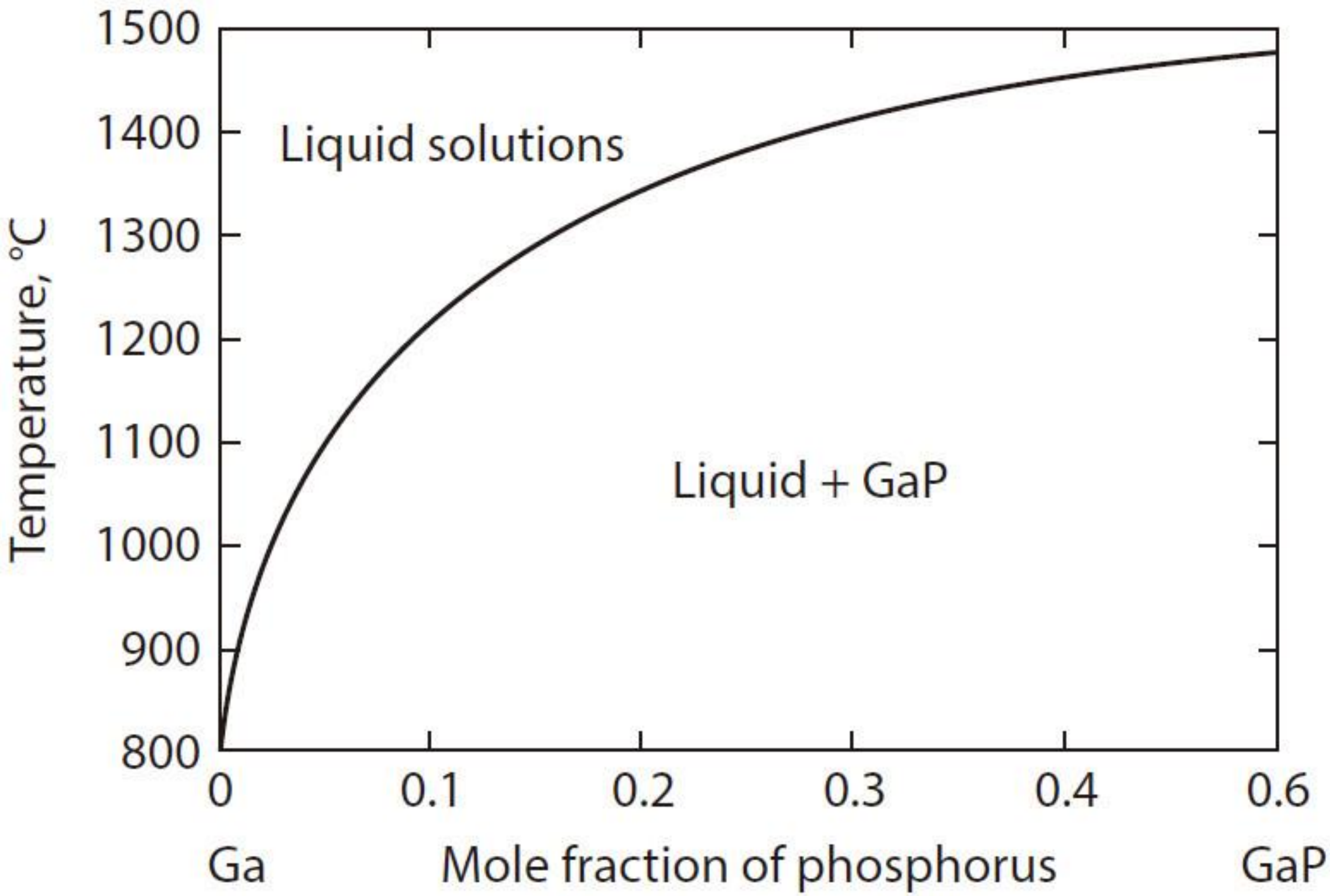


**Figure 13.13** The phase diagram for the system Ga–GaP.

**Solution**

At 1273 K,

$$\Delta G^{\circ}_{1273\ \mathrm{K}} = -3.968\times10^{4}\ \mathrm{J} = -8.3144\times1273\ln K_{1273\ \mathrm{K}}$$

which gives

$$K_{1273\ \mathrm{K}} = 24.97 = \frac{a_{\mathrm{GaP}}}{a_{\mathrm{Ga}}\, p_{\mathrm{P_2}}^{1/2}}$$

In the preceding expression, $a_{\mathrm{Ga}}$ is the activity of Ga in the liquidus melt with respect to liquid Ga as the standard state, and, since the liquidus melt is in equilibrium with pure solid GaP, the activity of GaP, $a_{\mathrm{GaP}}$ , is unity.

**Solution**

The variation of the liquidus composition with temperature in the range 1173–1373 K can be expressed as

$$\ln X_{\mathrm{P}} = -\frac{16,550}{T} + 9.902$$

which gives the liquidus composition at 1273 K as $X_{\mathrm{P}} = 0.045$.

In view of the low solubility of the solute P, it can be assumed that the solvent Ga obeys Raoult’ s law, in which case the activity of Ga in the liquidus melt is 0.955, and hence the partial pressure of $P_2$ exerted by the liquidus melt is

$$p_{\mathrm{P}_2} = \left(\frac{1}{24.97 \times 0.955}\right)^2 = 1.76 \times 10^{-3} \text{ atm}$$

**Example 2: The Mg–Si System**

The phase diagram for the system Mg–Si is shown in Figure 13.14. Determine the extent to which the phase diagram can be calculated, assuming that the liquid solutions exhibit regular solution behavior.

- Magnesium melts at 921 K and has a Gibbs free energy change on melting of $\Delta G^\circ_{m,\mathrm{Mg}} = 8790 - 9.54T$ J.
- Silicon melts at 1688 K and has a Gibbs free energy change on melting of $\Delta G^\circ_{m,\mathrm{Si}} = 50,630 - 30.0T$ J.
- $Mg_2Si$ melts at 1358 K and has a Gibbs free energy change on melting of $\Delta G^\circ_{m,\mathrm{Mg_2Si}} = 85,770 - 63.2T$ J.

The standard Gibbs free energy change for the reaction

$$2\mathrm{Mg}_{(l)} + \mathrm{Si}_{(s)} = \mathrm{Mg_2Si}_{(s)} \qquad \text{(i)}$$

is

$$\Delta G^\circ_{(\mathrm{i})} = -100,400 + 39.3T \text{ J}$$

# 13.6 Binary Systems Containing Compounds

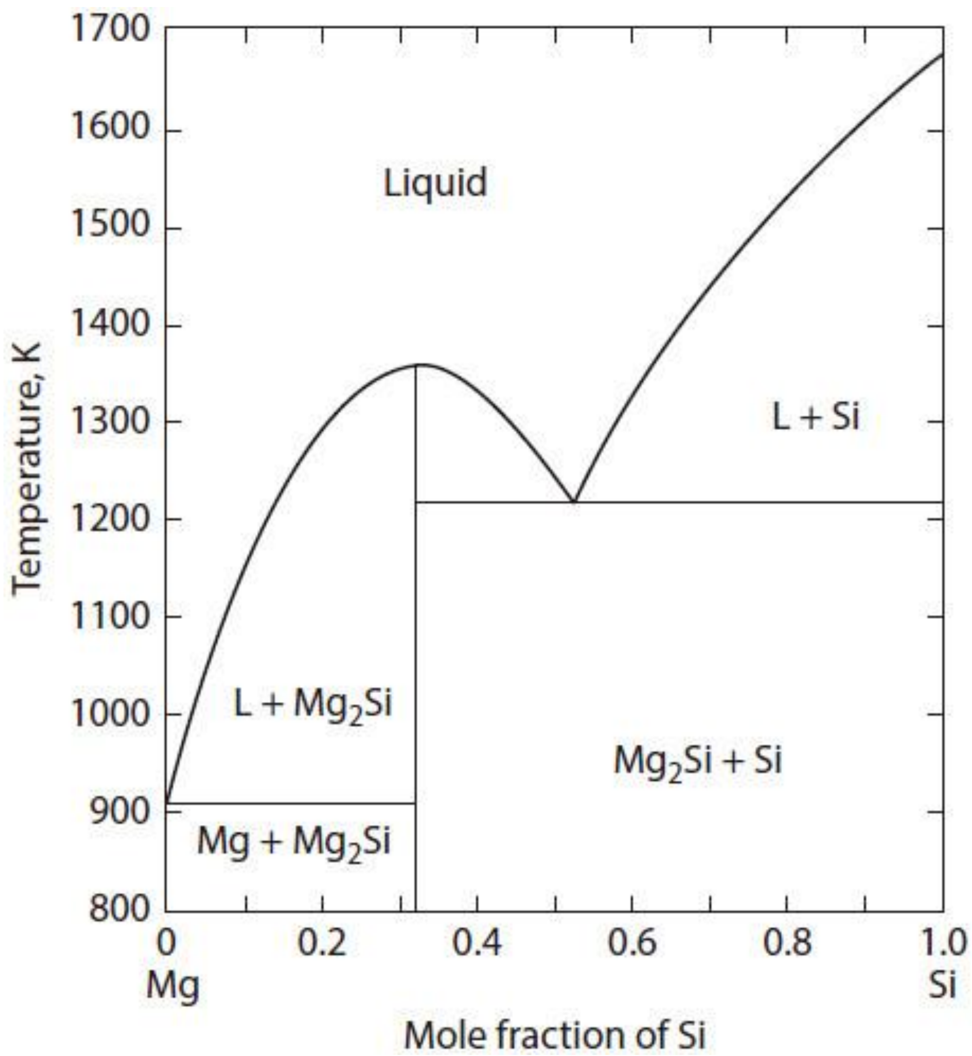


**Figure 13.14** The phase diagram for the system Mg–Si.

**Solution**

- ✓ The Gibbs free energy diagram at 1358 K for the system, using liquid as the standard state for Mg and solid as the standard state for Si, is shown in Figure 13.15.
- ✓ $\Delta G^\circ_{(\mathrm{i})} = -47,030\ \mathrm{J}$ at the melting temperature of $Mg_2Si$ (1358 K), and thus, the Gibbs free energy of formation of $Mg_{2/3}Si_{1/3}$ $= -47,030/3 = -15,676\ \mathrm{J}$, and this is the length of the line *de* in Figure 13.15.
- ✓ Point *b* in Figure 13.15 represents the free energy of liquid Si relative to solid Si and lies 9890 J above the point *a*.
- ✓ Consequently, the length of the line *cd* is 9890/3 = 3297 J, and the length of the line *ce* is 3,297 + 15,676 = 18,973 J. Thus, the Gibbs free energy of formation of solid $Mg_{2/3}Si_{1/3}$ from liquid Mg and liquid Si is –18,919 J.

# 13.6 Binary Systems Containing Compounds

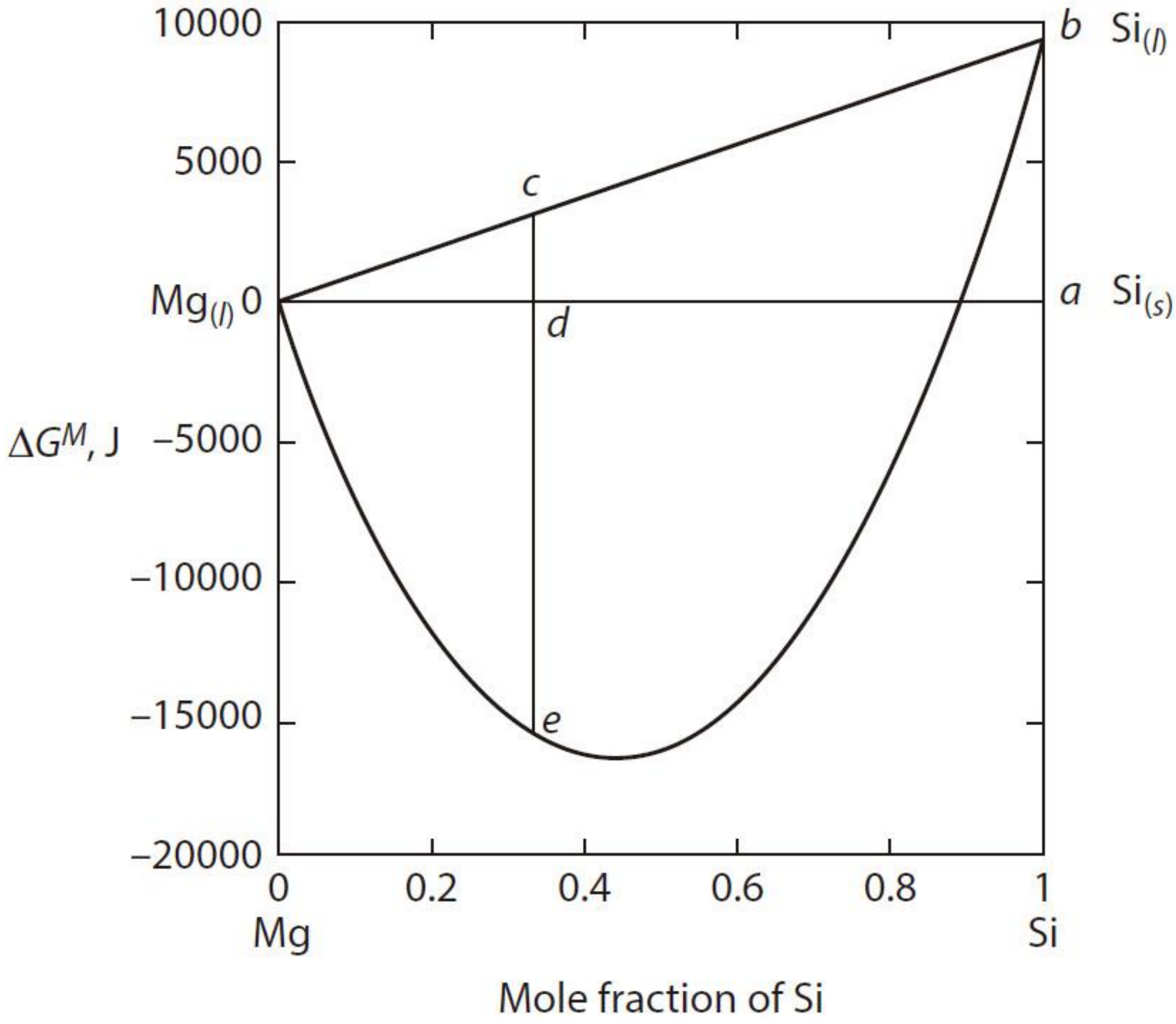


**Figure 13.15** Molar Gibbs free energies of mixing in the system Mg–Si at 1358 K.

# 13.6 Binary Systems Containing Compounds

- ✓ However, at the melting temperature of 1358 K, $\Delta G^\circ_{\mathrm{Mg_2Si},(s)} = \Delta G^\circ_{\mathrm{Mg_2Si},(l)}$, and thus, the Gibbs free energy of formation of liquid $Mg_{2/3}Si_{1/3}$ from liquid Mg and liquid Si at 1358 K is also 973 J.
- ✓ Thus, the line representing the molar Gibbs free energy of formation of melts in the system at 1358 K passes through the point *e* and from the general expression for the formation of a regular solution

$$\Delta G^M = RT\left(X_{\mathrm{Mg}} \ln X_{\mathrm{Mg}} + X_{\mathrm{Si}} \ln X_{\mathrm{Si}}\right) + \alpha X_{\mathrm{Mg}} X_{\mathrm{Si}}$$

at $X_{\mathrm{Si}} = 1/3$ ,

$$-18{,}973 = 8.3144 \times 1358 \left(\frac{2}{3}\ln\frac{2}{3} + \frac{1}{3}\ln\frac{1}{3}\right) + \alpha \cdot \frac{1}{3} \cdot \frac{2}{3}$$

which gives α = –53,040 J.

# 13.6 Binary Systems Containing Compounds

- Combination of $\Delta G^\circ_{(\text{i})}$ and the Gibbs free energy change for the melting of Si gives

$$\Delta G^\circ_{(\text{ii})} = -151,030 + 69.3T \text{ J}$$

for the reaction

$$2\text{Mg}_{(l)} + \text{Si}_{(l)} = \text{Mg}_2\text{Si}_{(s)} \qquad \text{(ii)}$$

- Thus,

$$-151,030 + 69.3T = -RT\ln K = -RT\ln\frac{a_{\text{Mg}_2\text{Si},(s)}}{a^2_{\text{Mg}}a_{\text{Si}}}$$

- Since liquids on the $Mg_2Si$ liquidus line are saturated with $Mg_2Si$, $a_{\text{Mg}_2\text{Si},(s)} = 1$ , the variations of the activities of Mg and Si with temperature along the $Mg_2Si$ liquidus line are given by

$$-15,030 + 69.3T = 2RT\ln a_{\text{Mg}} + RT\ln a_{\text{Si}} \qquad \text{(iii)}$$

# 13.6 Binary Systems Containing Compounds

- In a regular solution,

$$RT\ln a_i = RT\ln X_i + \Omega\left(1 - X_i\right)^2$$

and thus, Equation (iii) becomes

$$\begin{aligned}-15{,}030 + 69.3T = {} & 2RT\ln\left(1 - X_{\mathrm{Si}}\right) - 2\times 53{,}040\left(1 - X_{\mathrm{Si}}\right)^2 \\ & + RT\ln X_{\mathrm{Si}} - 53{,}040\left(1 - X_{\mathrm{Si}}\right)^2\end{aligned} \qquad \text{(iv)}$$

- Equation (iv), which is the equation of the $Mg_2Si$ liquidus, is quadratic and gives two values of $X_{\mathrm{Si}}$ at each temperature, with one being the liquidus composition in the Mg–$Mg_2Si$ sub-binary and the other being the liquidus composition in the $Mg_2Si$– Si sub-binary.
- Equation (iv) is drawn as the broken line *abc* in Figure 13.16.

# 13.6 Binary Systems Containing Compounds

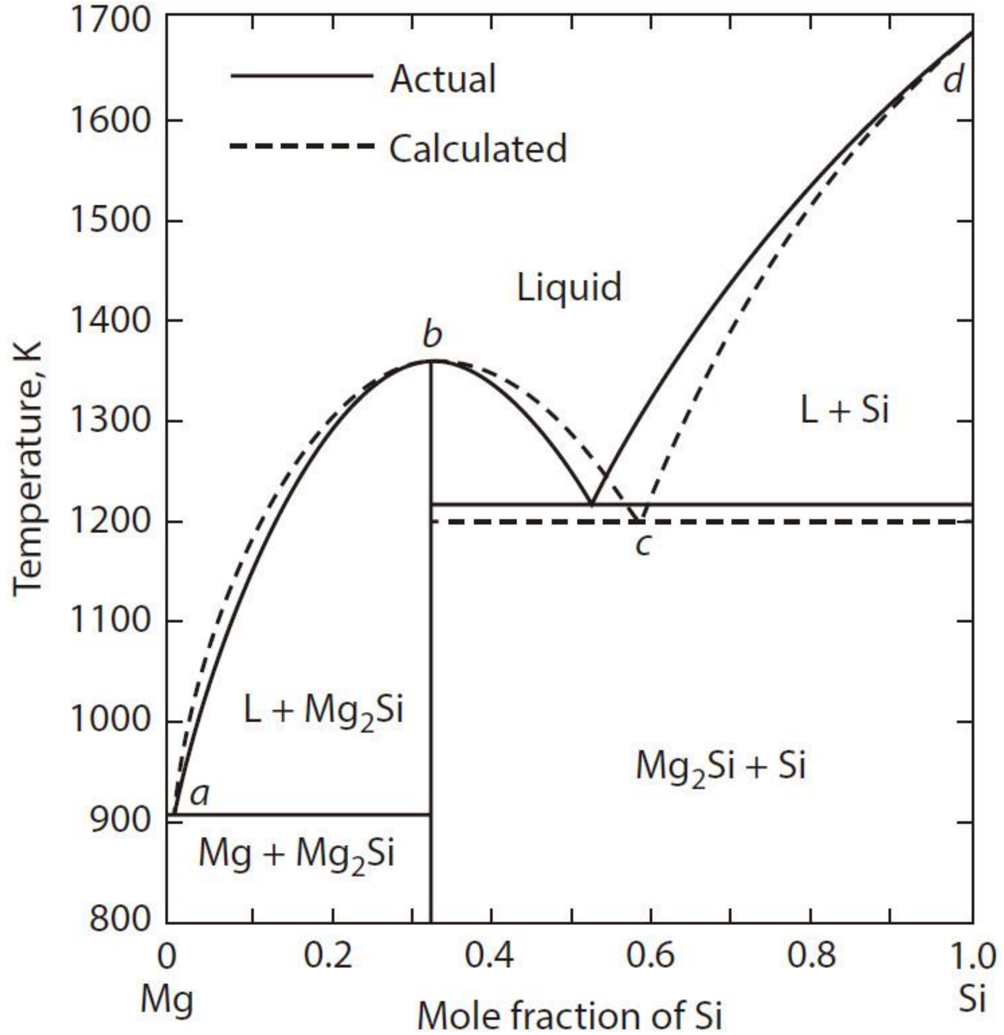


**Figure 13.16** Comparison between the calculated and the actual phase diagram for the system Mg–Si.

# 13.6 Binary Systems Containing Compounds

- The Si liquidus line is obtained from Equation 10.22 as

$$\Delta \overline{G}^{M}_{\mathrm{Si}(l)} = -\Delta G^{\circ}_{m,\mathrm{Si}}$$

- That is,

$$RT \ln X_{\mathrm{Si}} + \alpha \left(1 - X_{\mathrm{Si}}\right)^2 = -50,630 + 30.0T$$

which gives

$$T = \frac{50,630 - 53,040\left(1 - X_{\mathrm{Si}}\right)^2}{30.0 - 8.3144 \ln X_{\mathrm{Si}}} \qquad \text{(v)}$$

- Equation (v) is drawn as the broken line *cd* in Figure 13.16.

# 13.6 Binary Systems Containing Compounds

- ✓ Similarly, the Mg liquidus line is given by

$$RT\ln\left(1-X_{\mathrm{Si}}\right)+\alpha X_{\mathrm{Si}}{}^{2}=-8790+9.52T$$

which gives

$$T=\frac{8790-53,040X_{\mathrm{Si}}{}^{2}}{9.52-8.3144\ln\left(1-X_{\mathrm{Si}}\right)} \qquad \text{(vi)}$$

- ✓ The calculated diagram shows good agreement with the actual diagram; the calculated eutectic temperature and eutectic composition in the $Mg_2Si$–Si sub-binary are, respectively, 1200 K and $X_{\mathrm{Si}}$ = 0.58, which are close to the actual values of 1218 K and $X_{\mathrm{Si}}$ = 0.53, and the eutectic composition and temperature in the Mg–$Mg_2Si$ sub-binary coincide with the actual values.

# Homework

**1.** A $CH_4$–$H_2$ gas mixture at 1 atm total pressure, in which $p_{H_2} = 0.957$ atm, is equilibrated with an Fe–C alloy at 1000 K. Calculate the activity of C with respect to graphite in the alloy. What would the value of $p_{H_2}$ in the gas mixture (at $P_{total}$ = 1 atm) have to be in order to saturate the Fe with graphite at 1000 K?

# Homework

**2.** Calculate the activity of FeO in an FeO–$Al_2O_3$–$SiO_2$ melt below which theFeO cannot be reduced to pure liquid iron by a CO–$CO_2$ mixture of $p_{CO}/p_{CO_2} = 10^5$ at 1600 ℃.

# Homework

**3.** A piece of iron is to be heat treated at 1000 K in a $CO$-$CO_2$-$H_2O$-$H_2$ gas mixture at 1 atm pressure. The gas mixture is produced by mixing $CO_2$ and $H_2$ and allowing the equilibrium $CO_2 + H_2 = CO + H_2O$ to establish. Calculate

(a) The minimum $H_2/CO_2$ ratio in the inlet gas which can be admitted to the furnace without oxidizing the iron,

(b) the activity of carbon (with respect to graphite) in the equilibrated gas of this initial minimum $H_2/CO_2$ ratio,

(c) the total pressure to which the equilibrated gas would have to be raised to saturate the iron with graphite at 1000 K, and

(d) the effect, on the partial pressure of oxygen in the equilibrated gas, of this increase in total pressure.

# *The End*